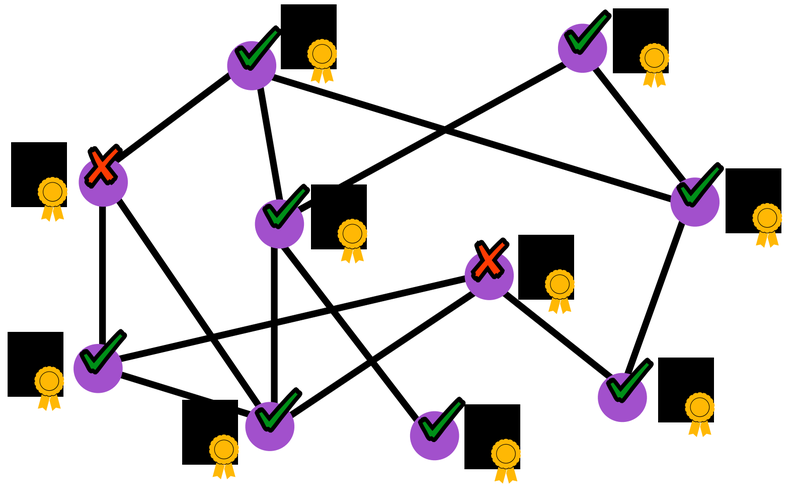
distributed SNARGs resilient to corrupt verifiers



Elette Boyle, Lali Devadas

NTT Research
Reichman University

The Cooper Union

this talk

this talk

- introduce ***distributed certification***

this talk

- introduce ***distributed certification***
- introduce our model: distributed SNARGs for **corrupted networks**

this talk

- introduce ***distributed certification***
- introduce our model: distributed SNARGs for **corrupted networks**
- state **our positive result**

this talk

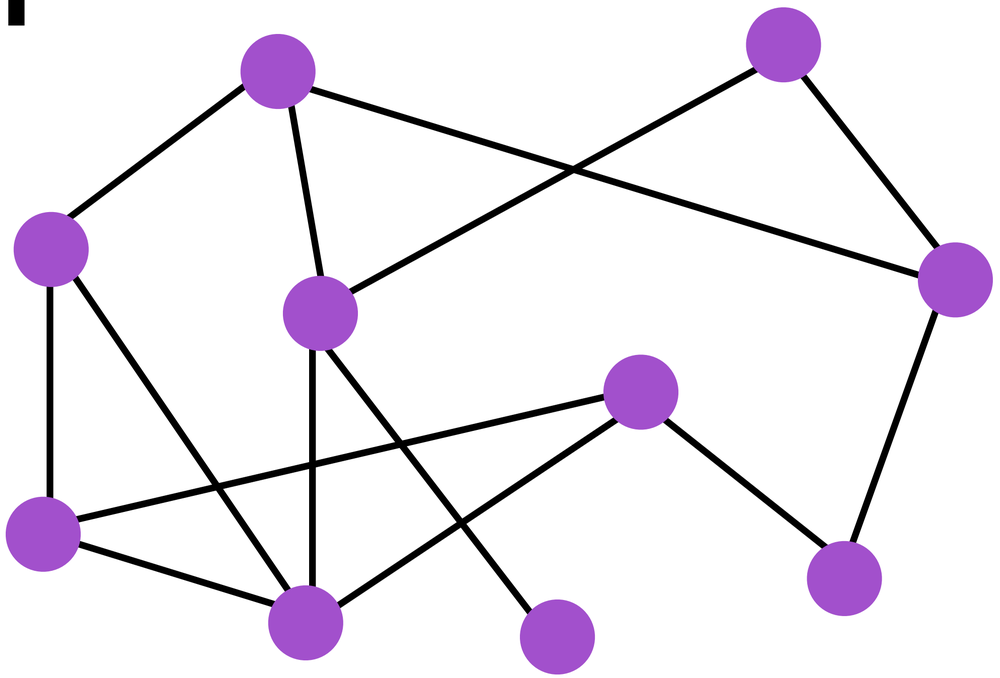
- introduce ***distributed certification***
- introduce our model: distributed SNARGs for **corrupted networks**
- state **our positive result**
- show a **lower bound**: achieving better succinctness likely requires heavier hammers

distributed certification

[KKP05,...]

no cryptography yet!

PROVER



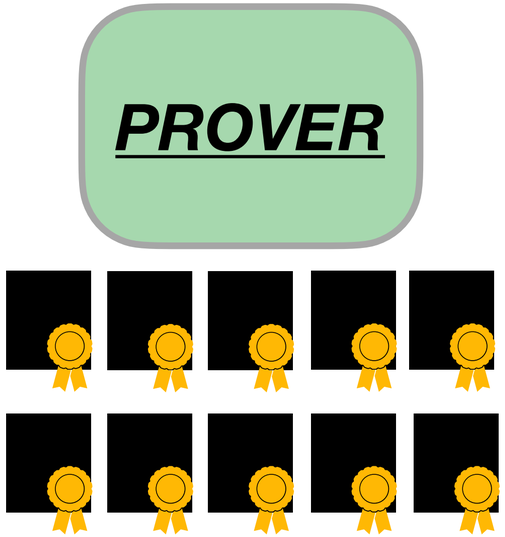
STATEMENT/VERIFIERS

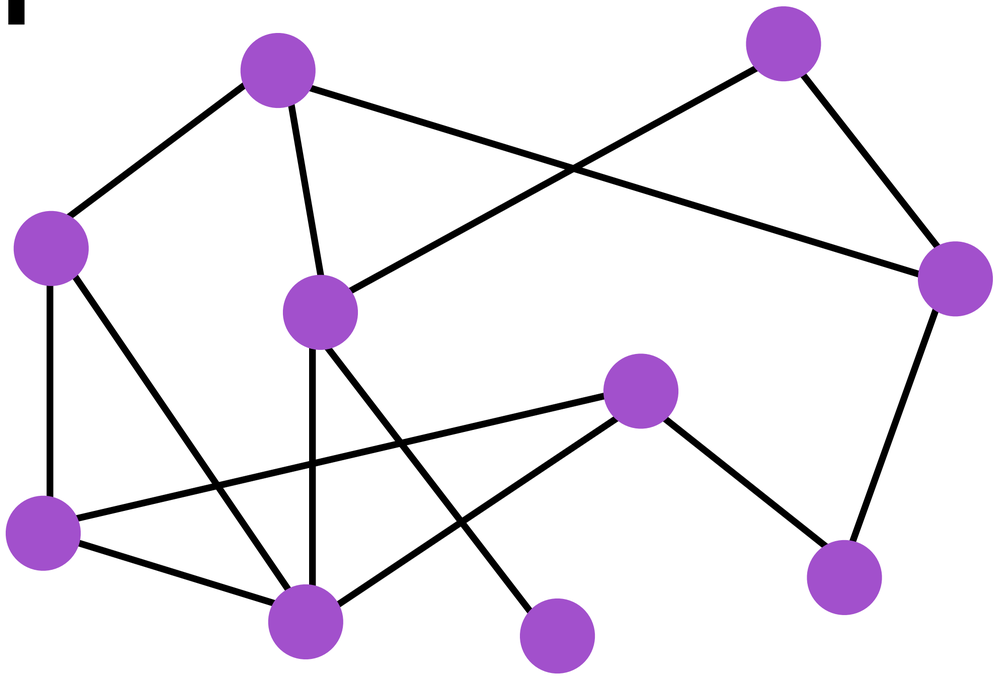
motivating applications include setting up network infrastructure and resource allocation

distributed certification

[KKP05,...]

no cryptography yet!



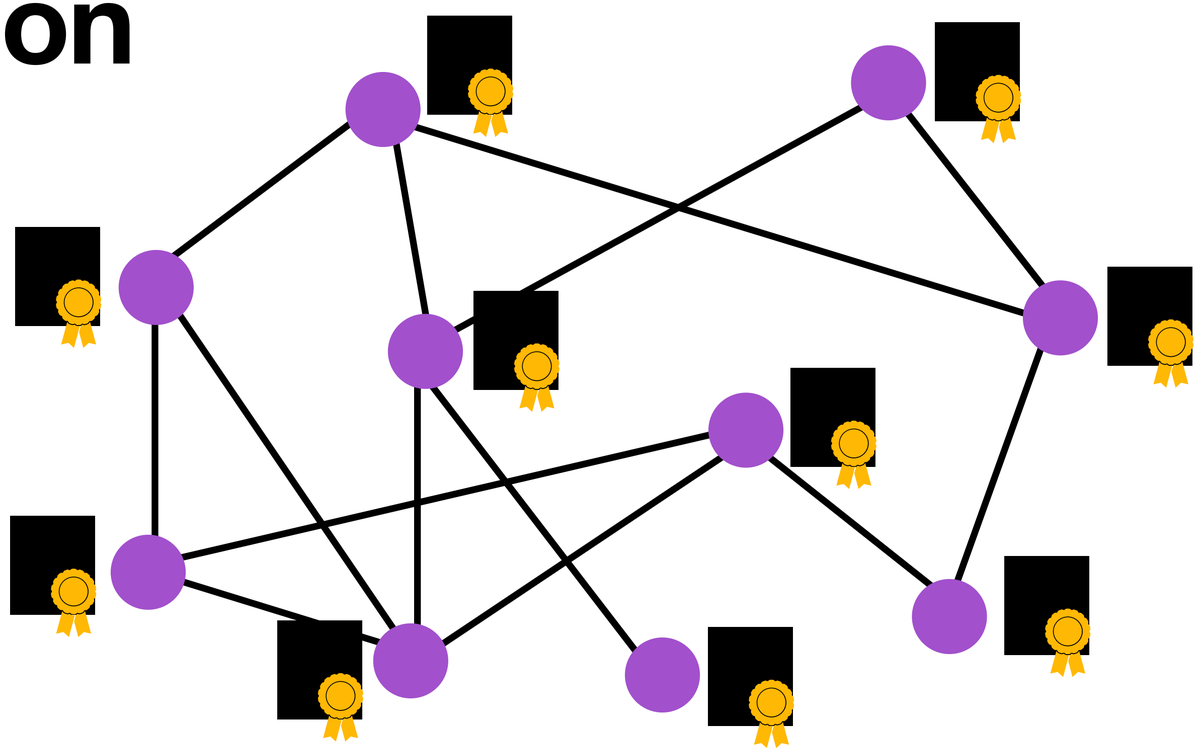


STATEMENT/VERIFIERS

motivating applications include setting up network infrastructure and resource allocation

distributed certification

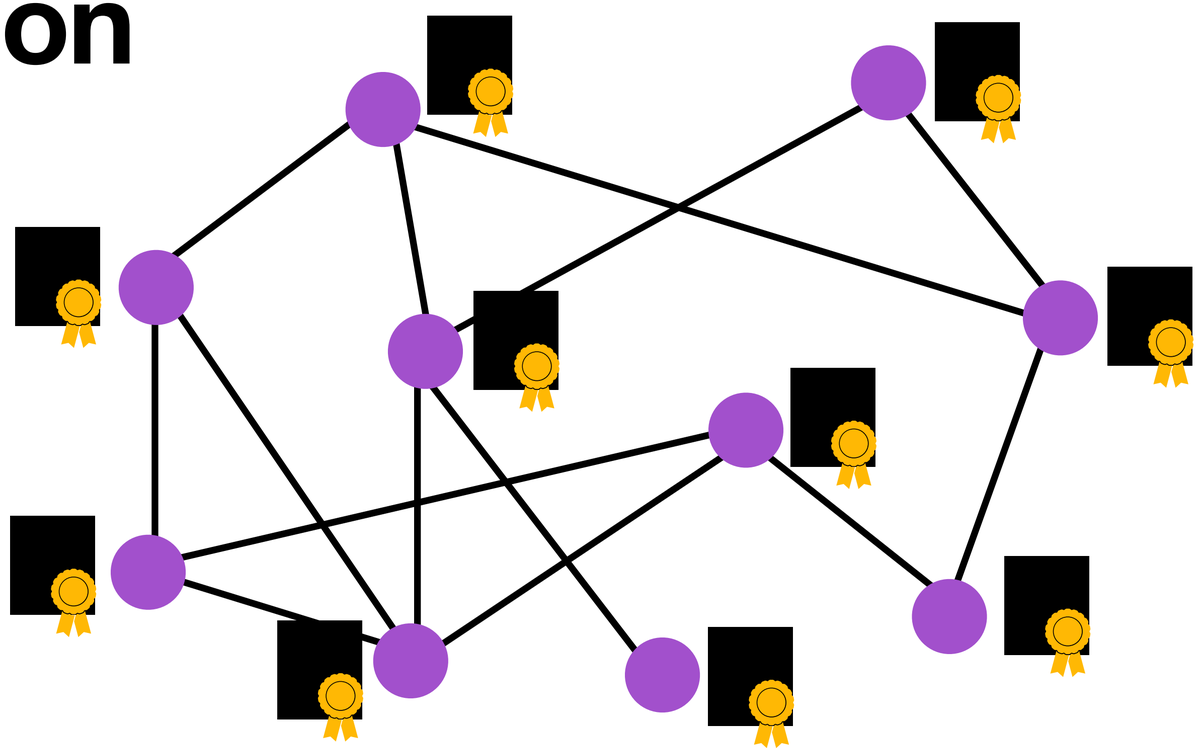
PROVER



STATEMENT/VERIFIERS

distributed certification

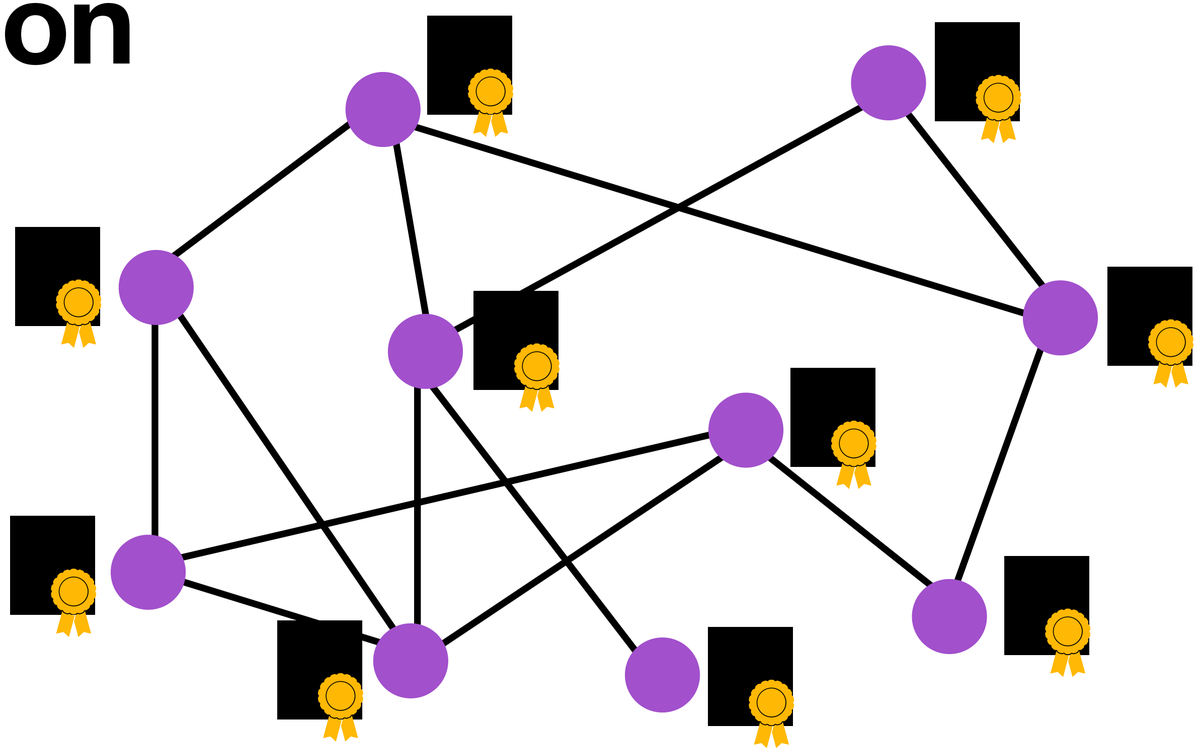
PROVER



STATEMENT/VERIFIERS

distributed certification

PROVER

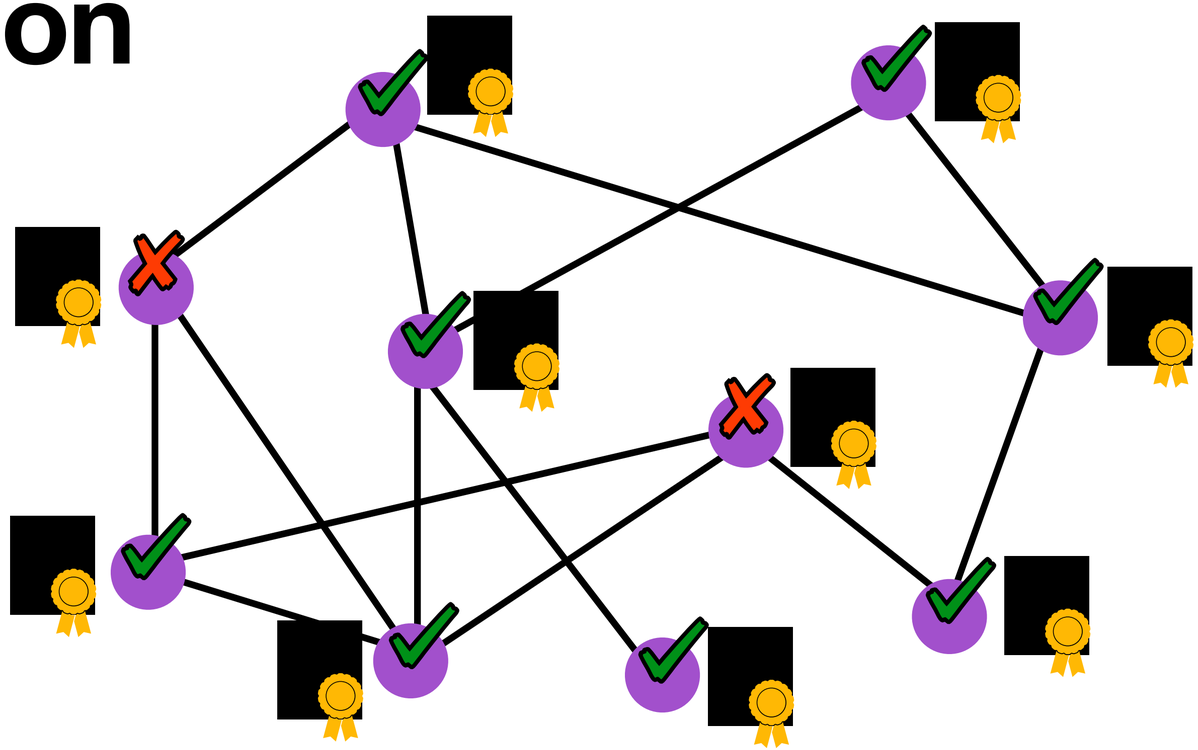


STATEMENT/VERIFIERS

n nodes only send each other *short* messages

distributed certification

PROVER

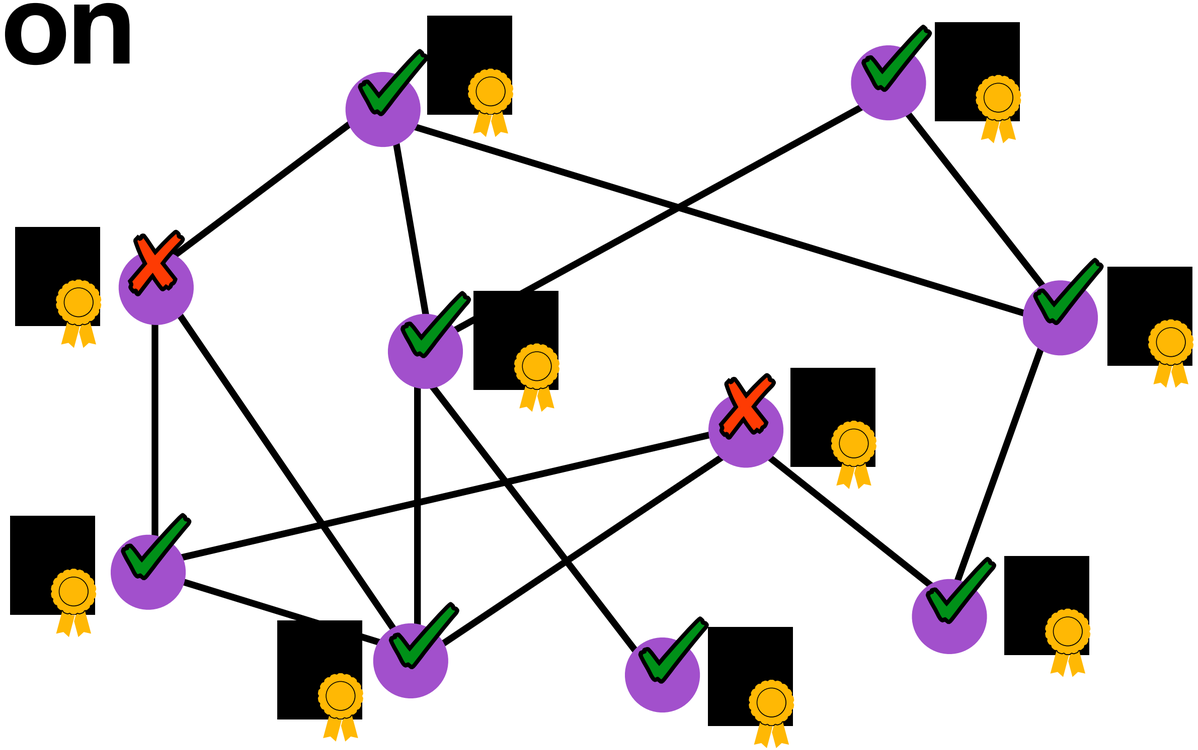


STATEMENT/VERIFIERS

n nodes only send each other *short* messages

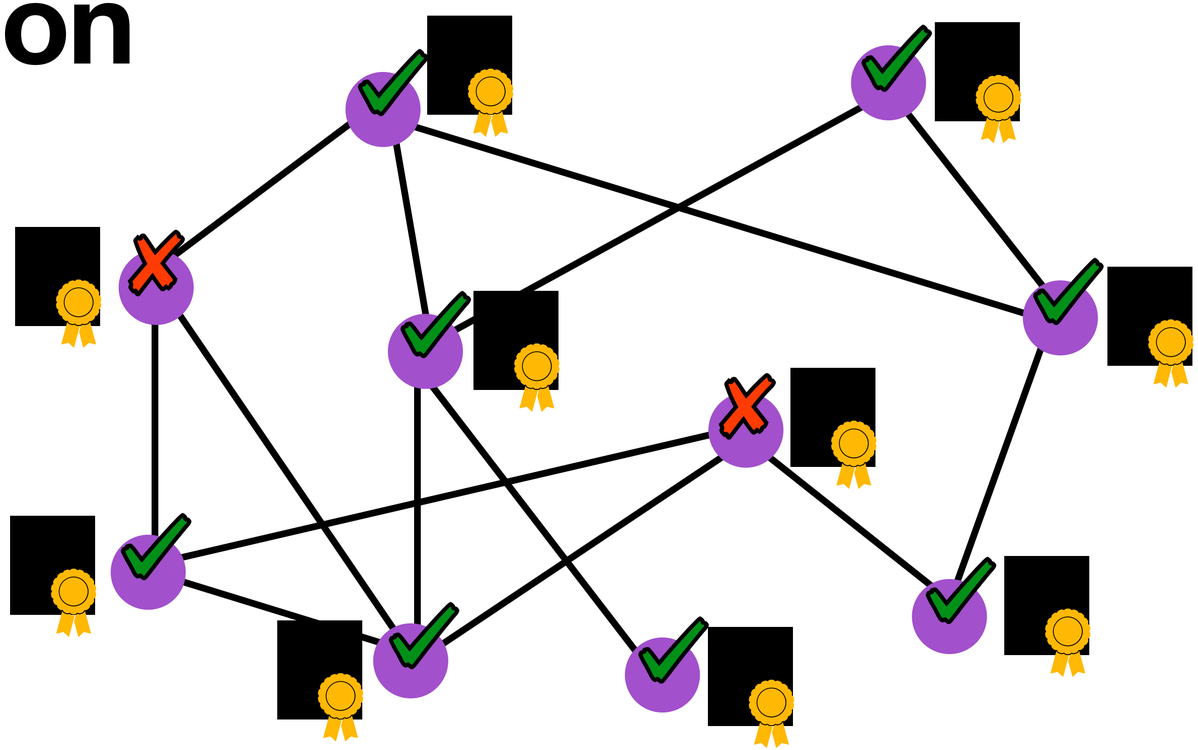
distributed certification

PROVER



distributed certification

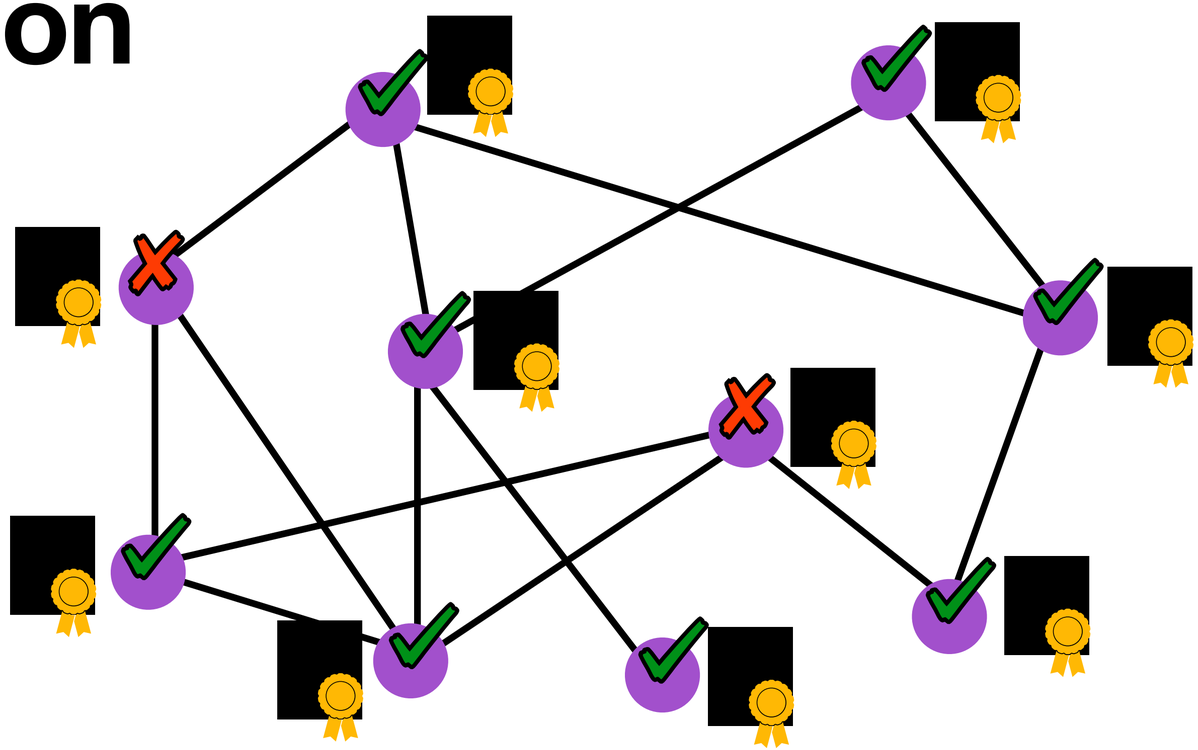
PROVER



COMPLETENESS if the network **satisfies** the property,
all verifiers accept honestly generated certificates

distributed certification

PROVER

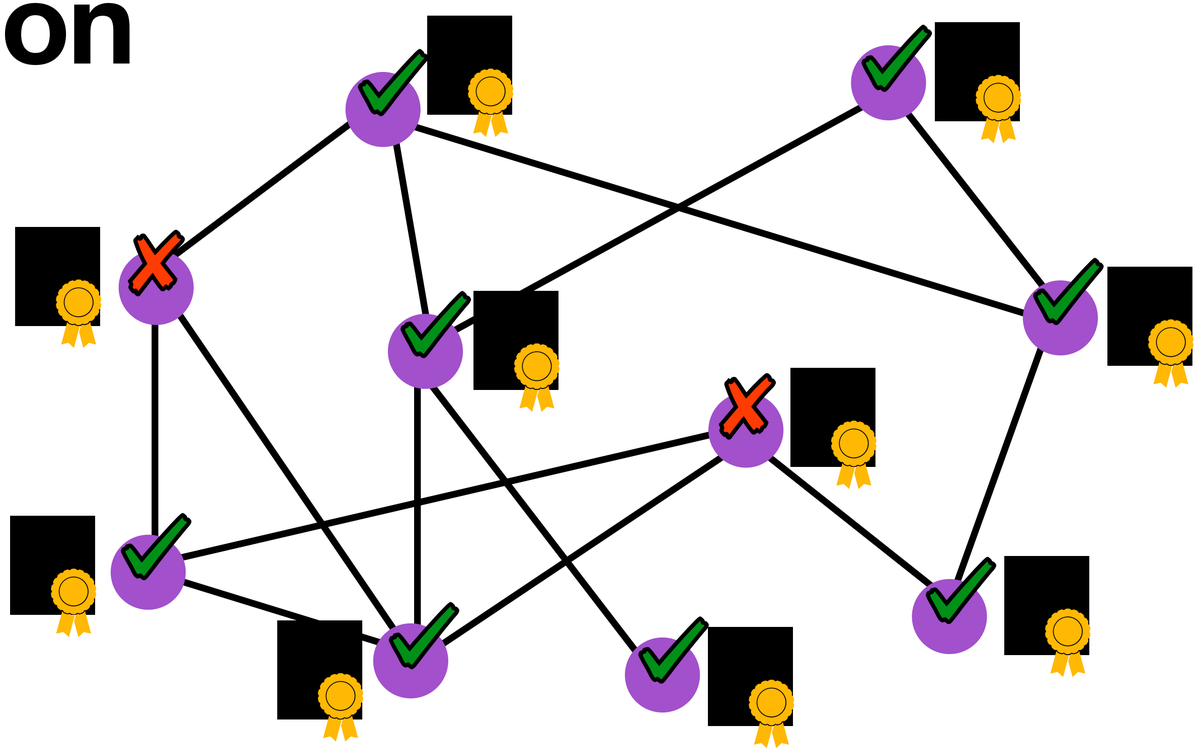


COMPLETENESS if the network **satisfies** the property, *all* verifiers accept honestly generated certificates

SOUNDNESS if the network **doesn't satisfy** the property, no cheating prover can trick *all* verifiers into accepting

distributed certification

PROVER

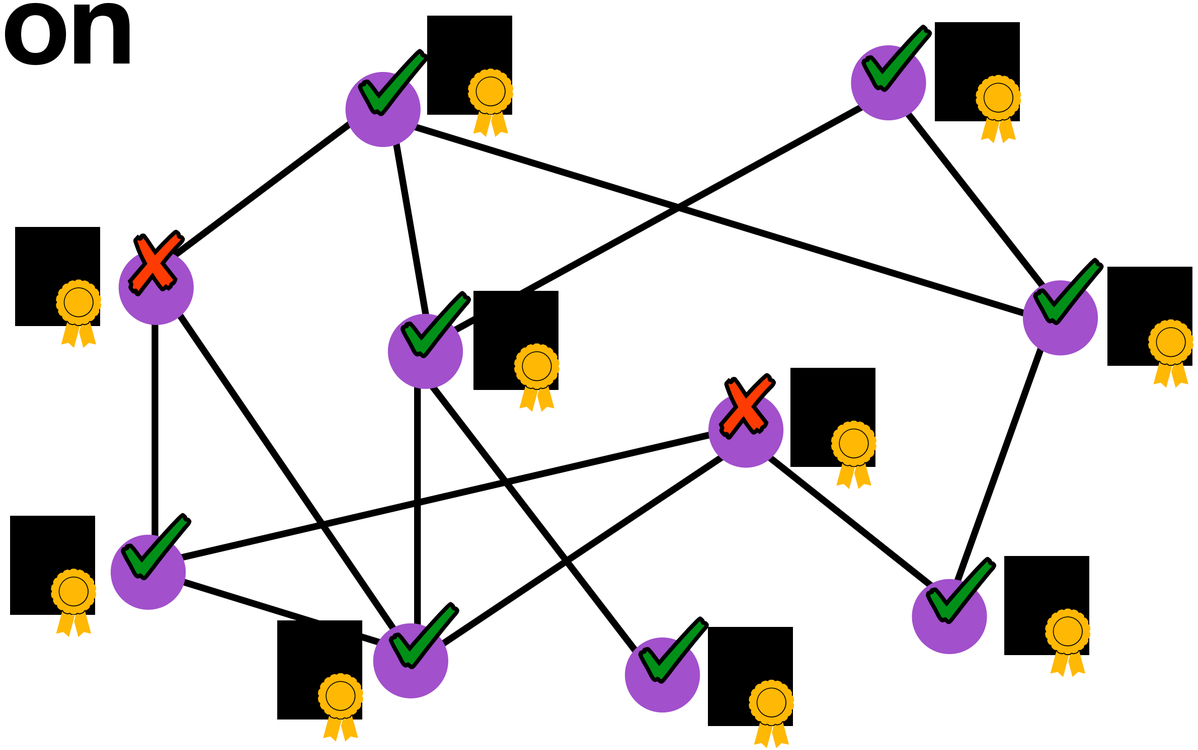


any property can be certified with certificates of length $O(|G|) = O(n^2)$ containing the network description

nodes exchange a collision-resistant hash of the network description to ensure they all received the same one

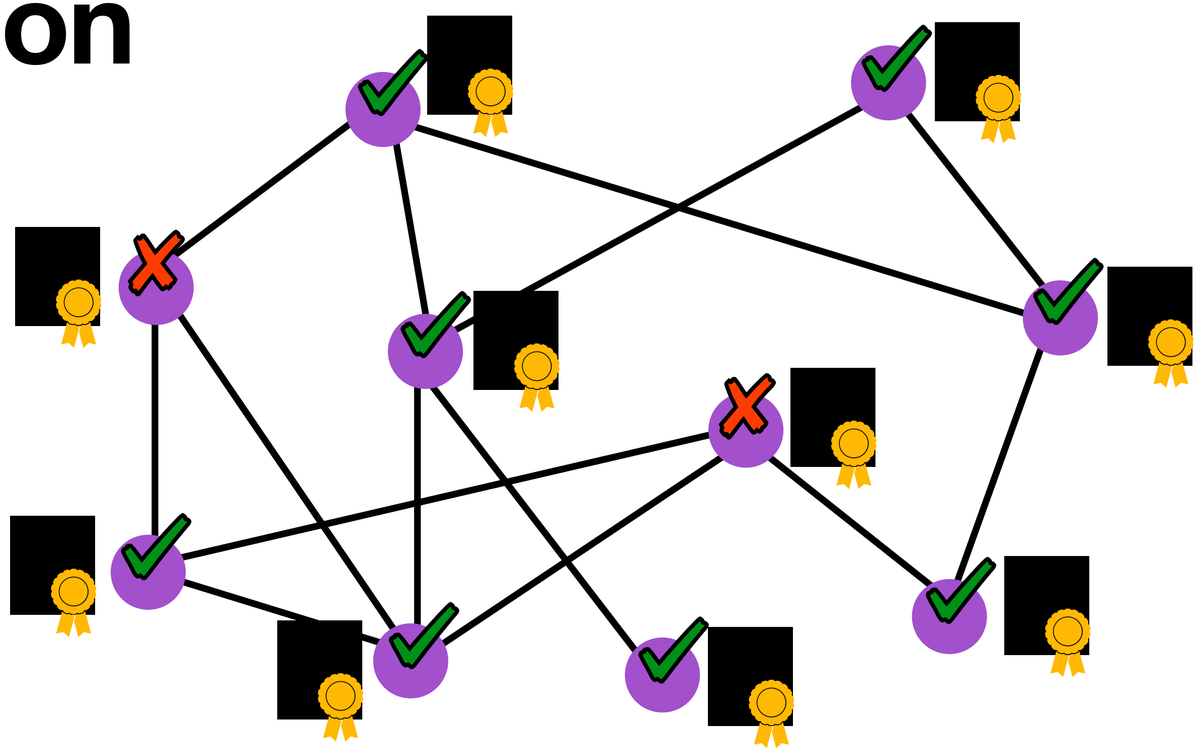
distributed certification

PROVER



distributed certification

PROVER

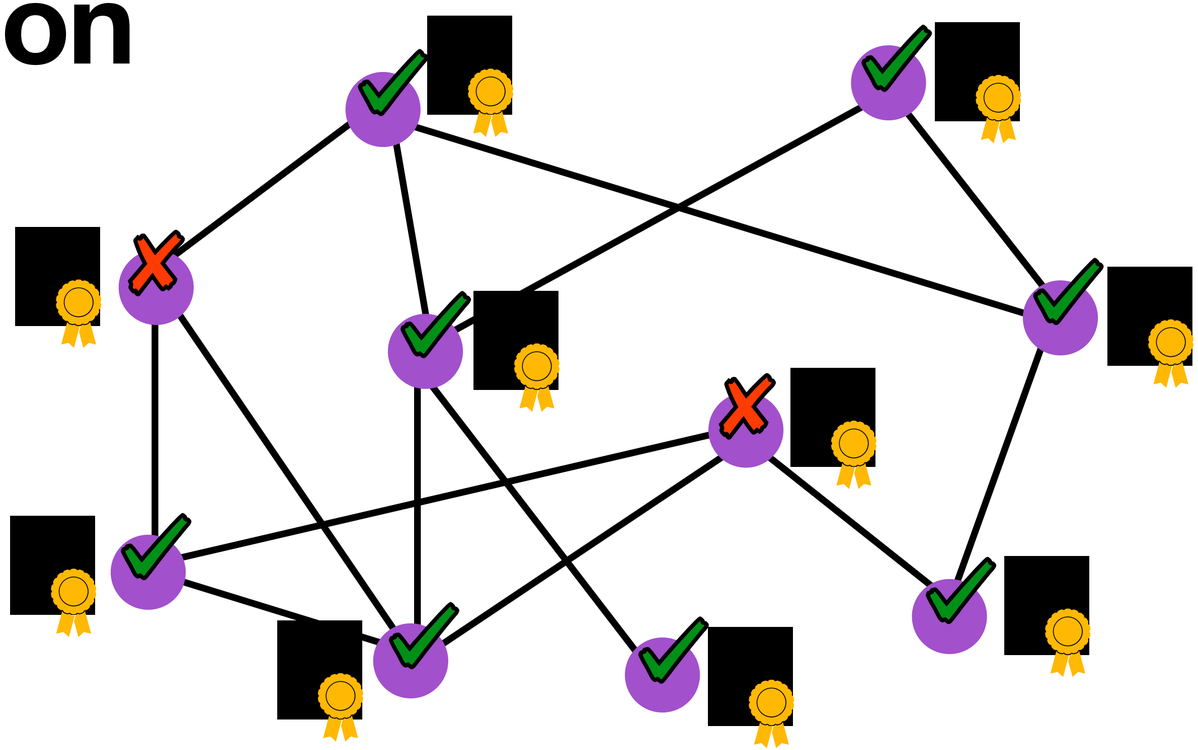


can this be succinct for all properties in P?

i.e., certificates asymptotically smaller than network description

distributed certification

PROVER



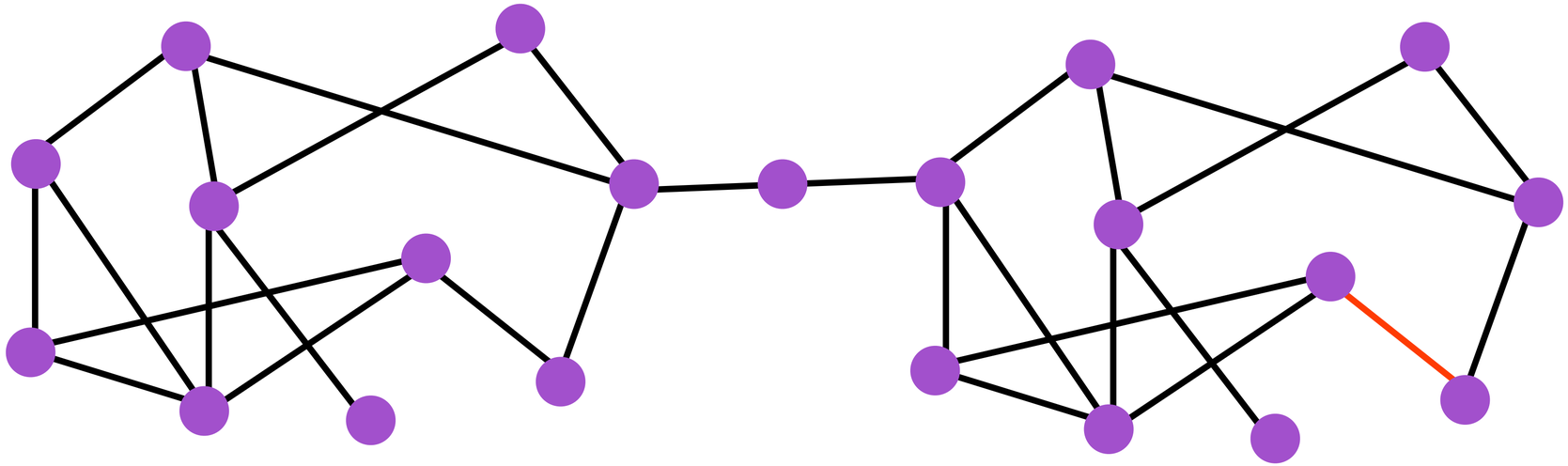
can this be succinct for all properties in P?

i.e., certificates asymptotically smaller than network description

[GS16, FMORT19]

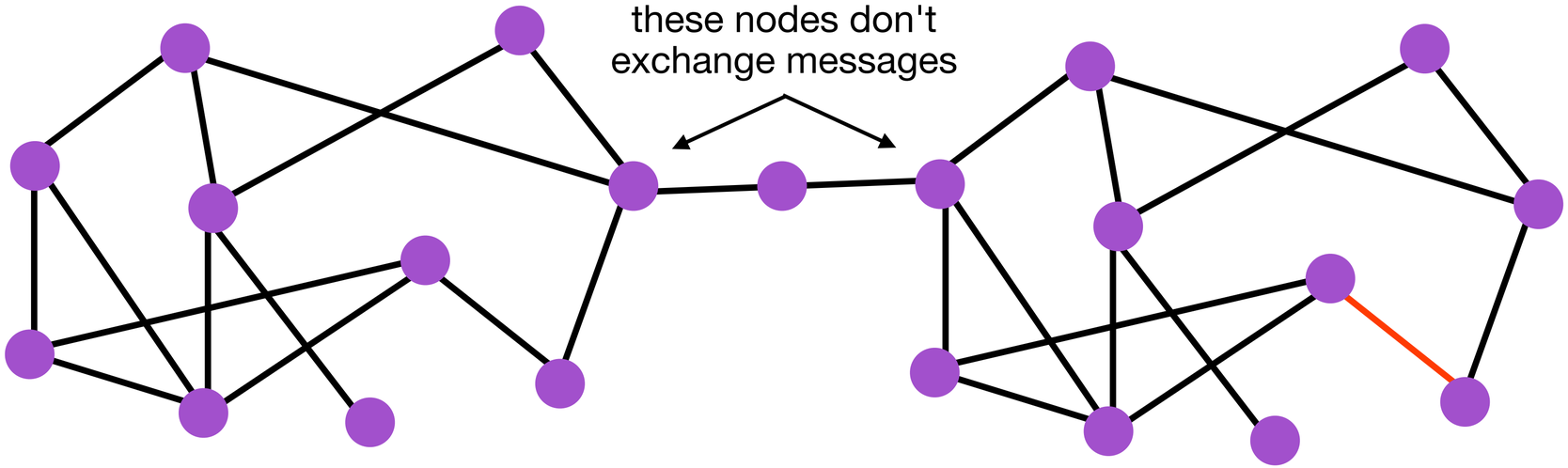
can distributed certification be succinct?

property: two copies of the same graph connected by a path



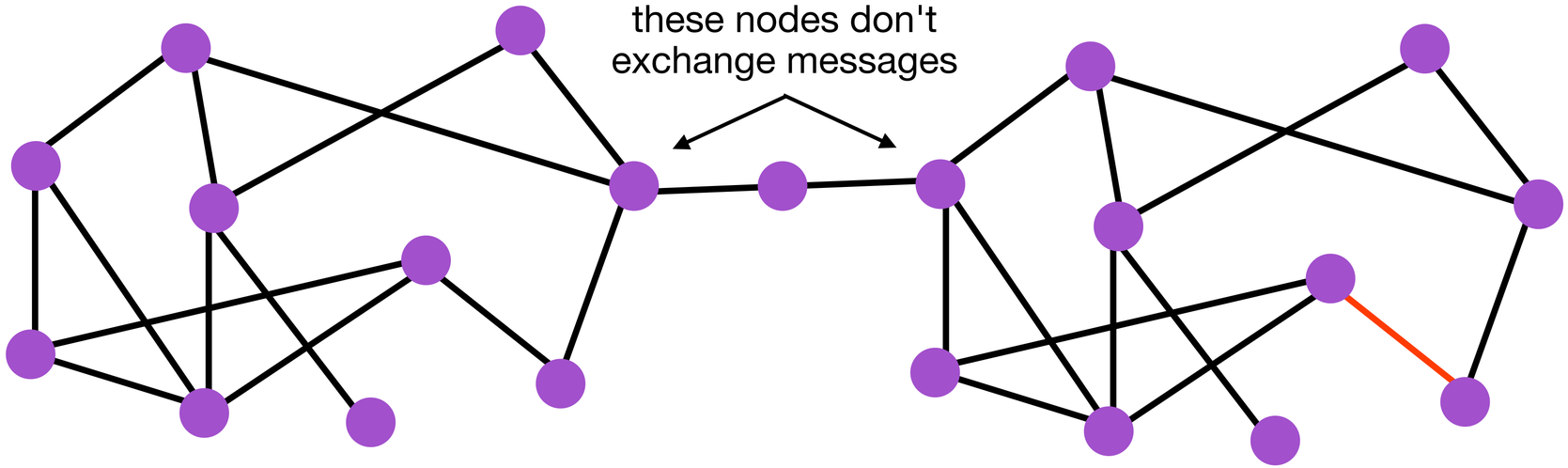
can distributed certification be succinct?

property: two copies of the same graph connected by a path



can distributed certification be succinct?

property: two copies of the same graph connected by a path

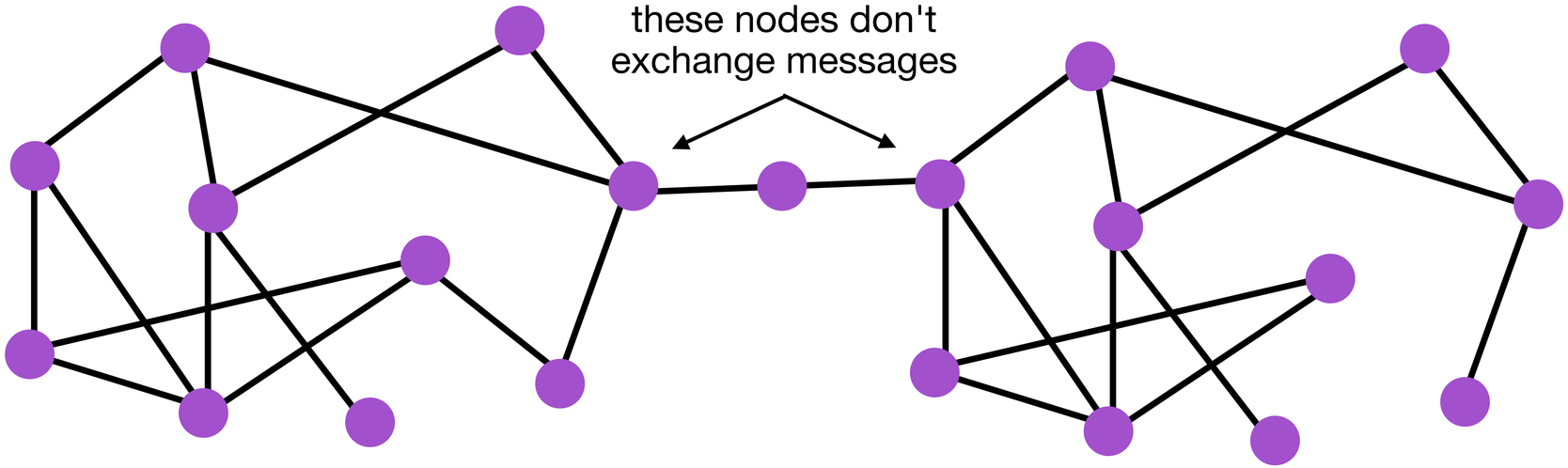


information cannot get from one side to another during verification

$\Longrightarrow$ **certificates need to encode the whole shape**

can distributed certification be succinct?

property: two copies of the same graph connected by a path

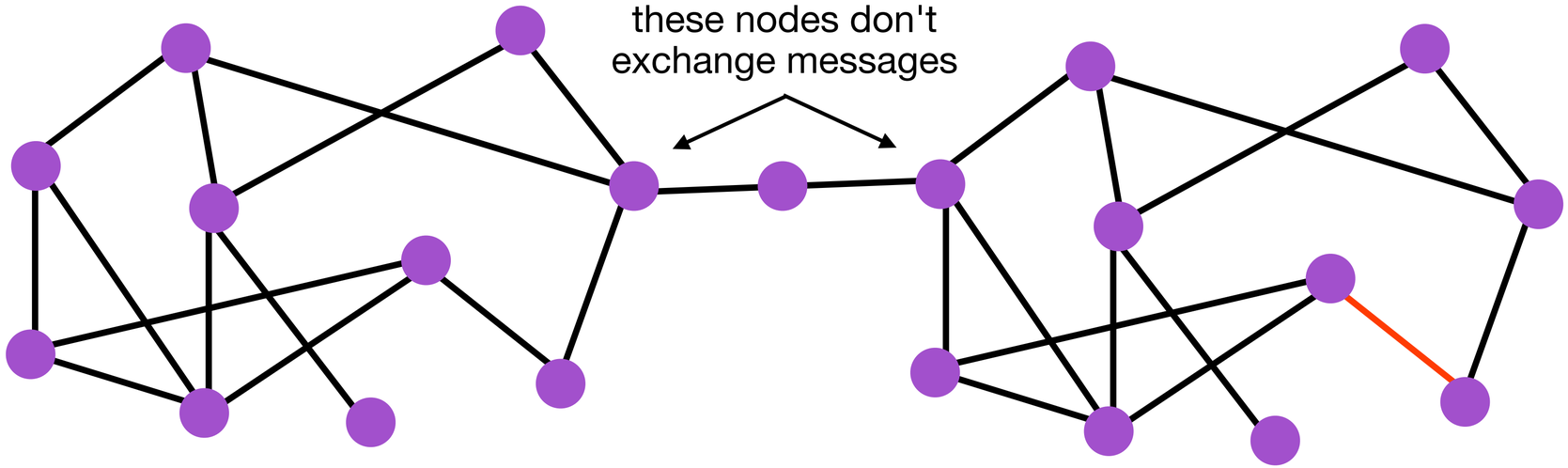


information cannot get from one side to another during verification

$\Longrightarrow$ **certificates need to encode the whole shape**

can distributed certification be succinct?

property: two copies of the same graph connected by a path



information cannot get from one side to another during verification

$\Longrightarrow$ **certificates need to encode the whole shape**

can distributed certification be *succinct*?

certificates of size $o(n^2)$



succinctness requires cryptography

we must impose a *computational bound* on the cheating prover

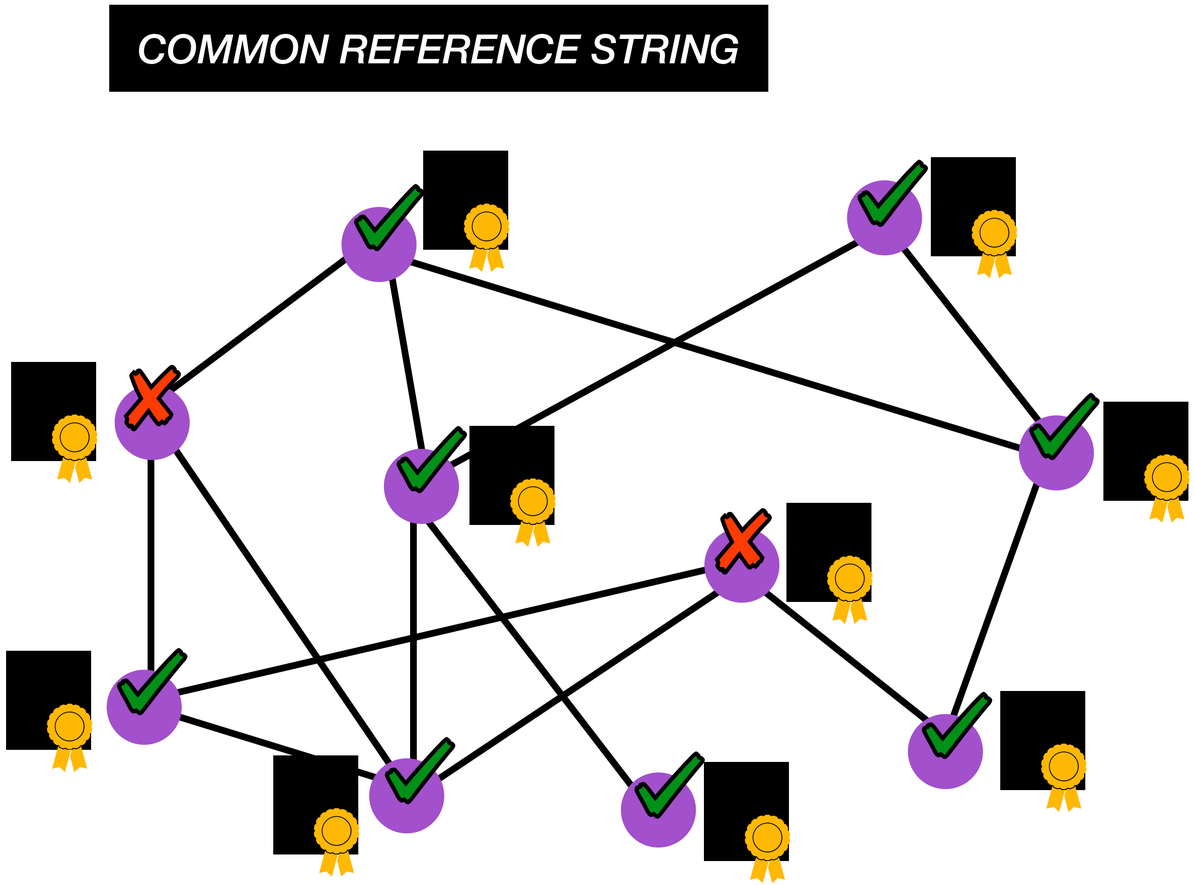
cryptography requires randomness

we must allow for a trusted *common reference string*

distributed SNARGs

[ABCMO23]

PROVER

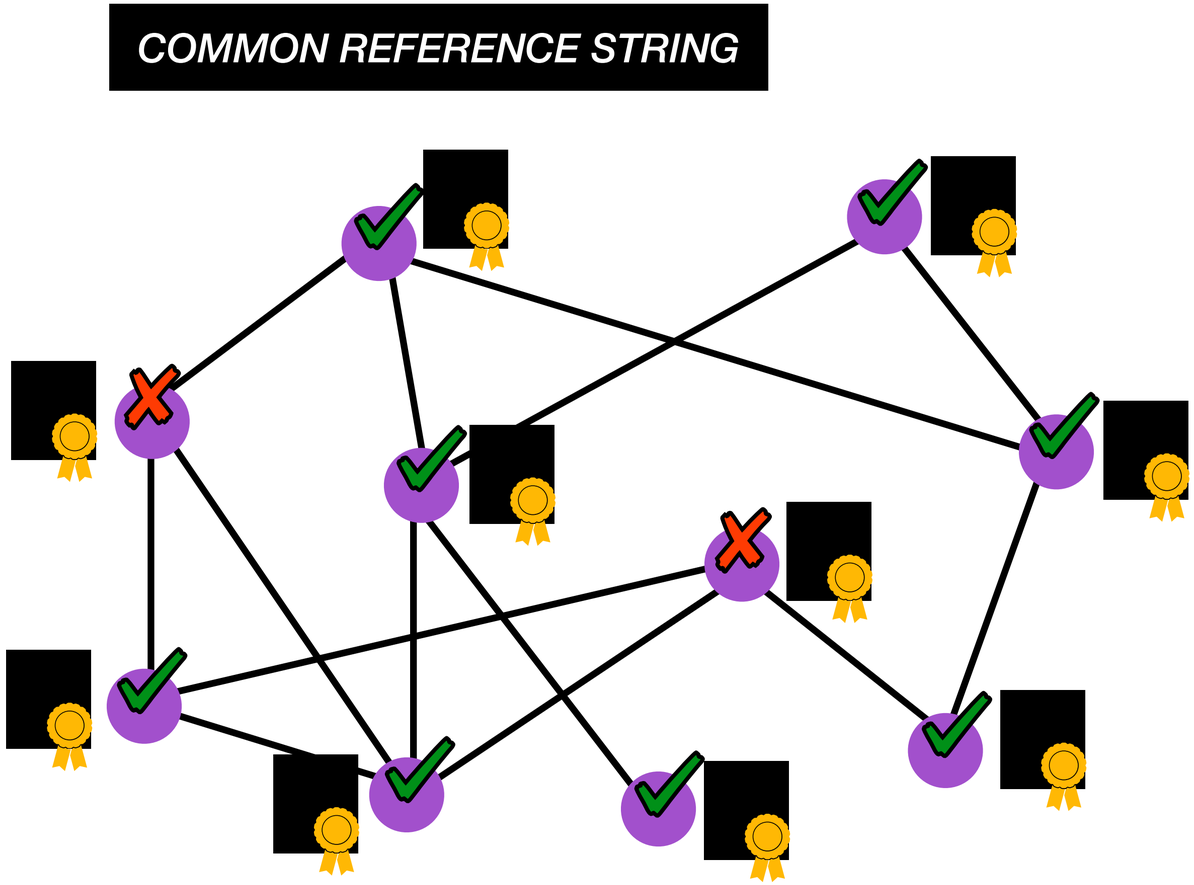


COMPLETENESS if the network **satisfies** the property, all verifiers accept honestly generated certificates

SOUNDNESS if the network **doesn't satisfy** the property, no efficient cheating prover can trick all verifiers into accepting

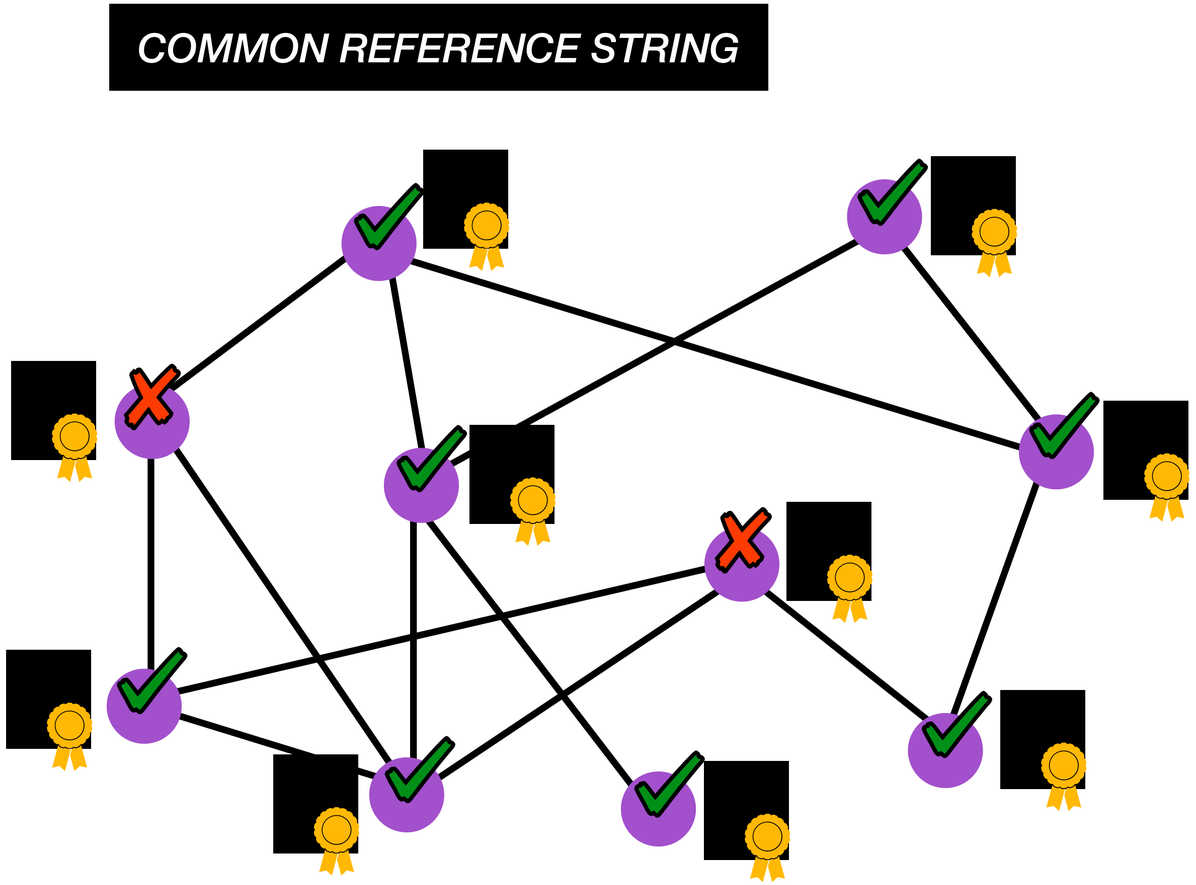
distributed SNARGs

PROVER



distributed SNARGs

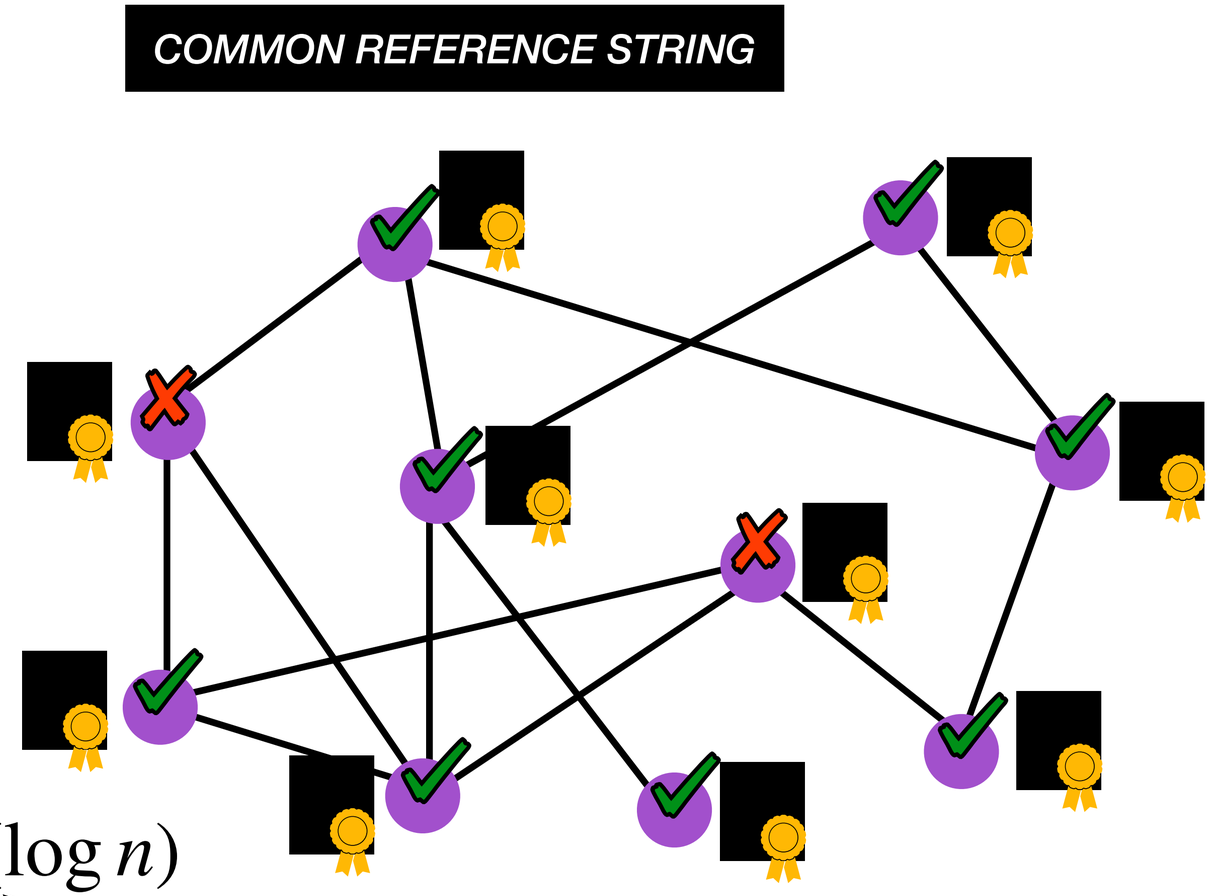
PROVER



[ABCMO23] constructs a distributed SNARG for any network property (in P)!

distributed SNARGs

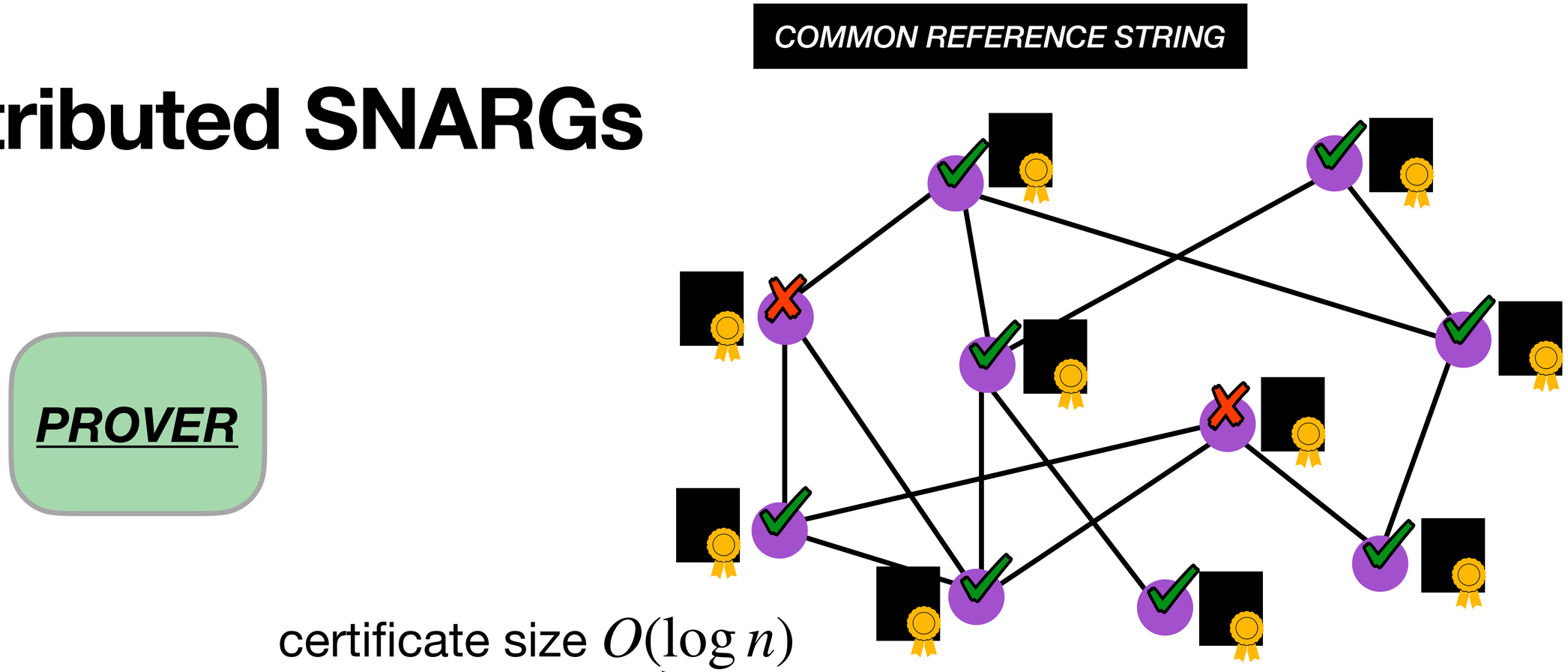
PROVER



certificate size $O(\log n)$

[ABCMO23] constructs a distributed SNARG for any network property (in P)!

distributed SNARGs



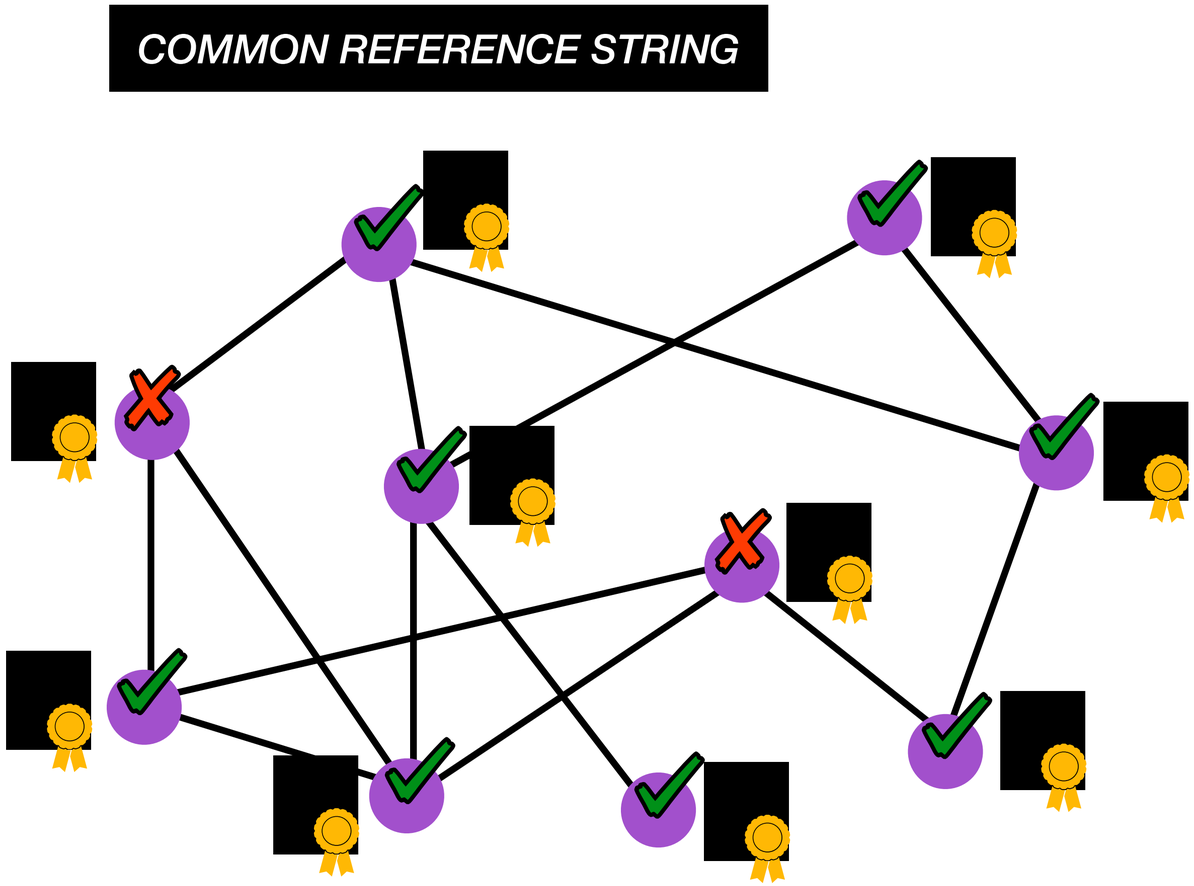
certificate size $O(\log n)$

[ABCMO23] constructs a distributed SNARG for any network property (in P)!

what if the adversary corrupts some nodes?

distributed SNARGs

PROVER

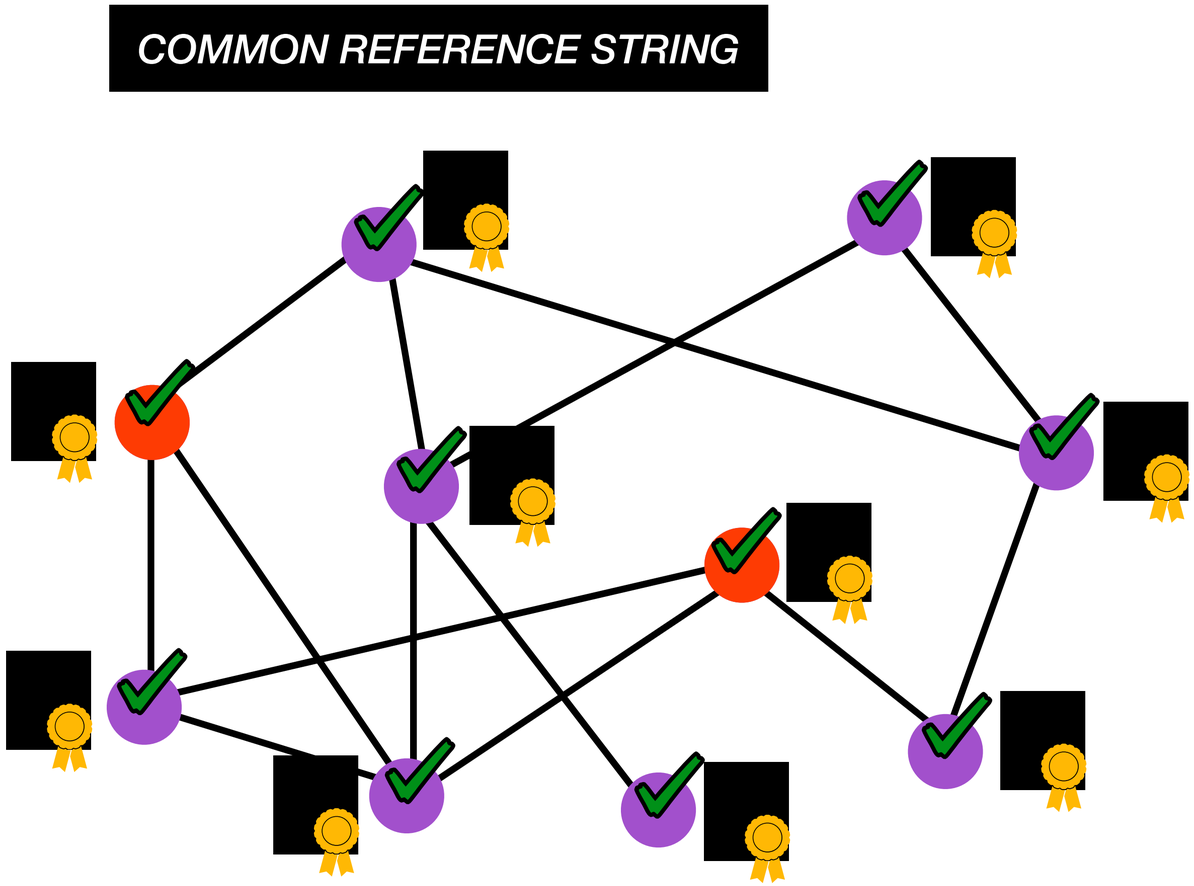


[ABCMO23] constructs a distributed SNARG for any network property (in P)!

what if the adversary corrupts some nodes?

distributed SNARGs

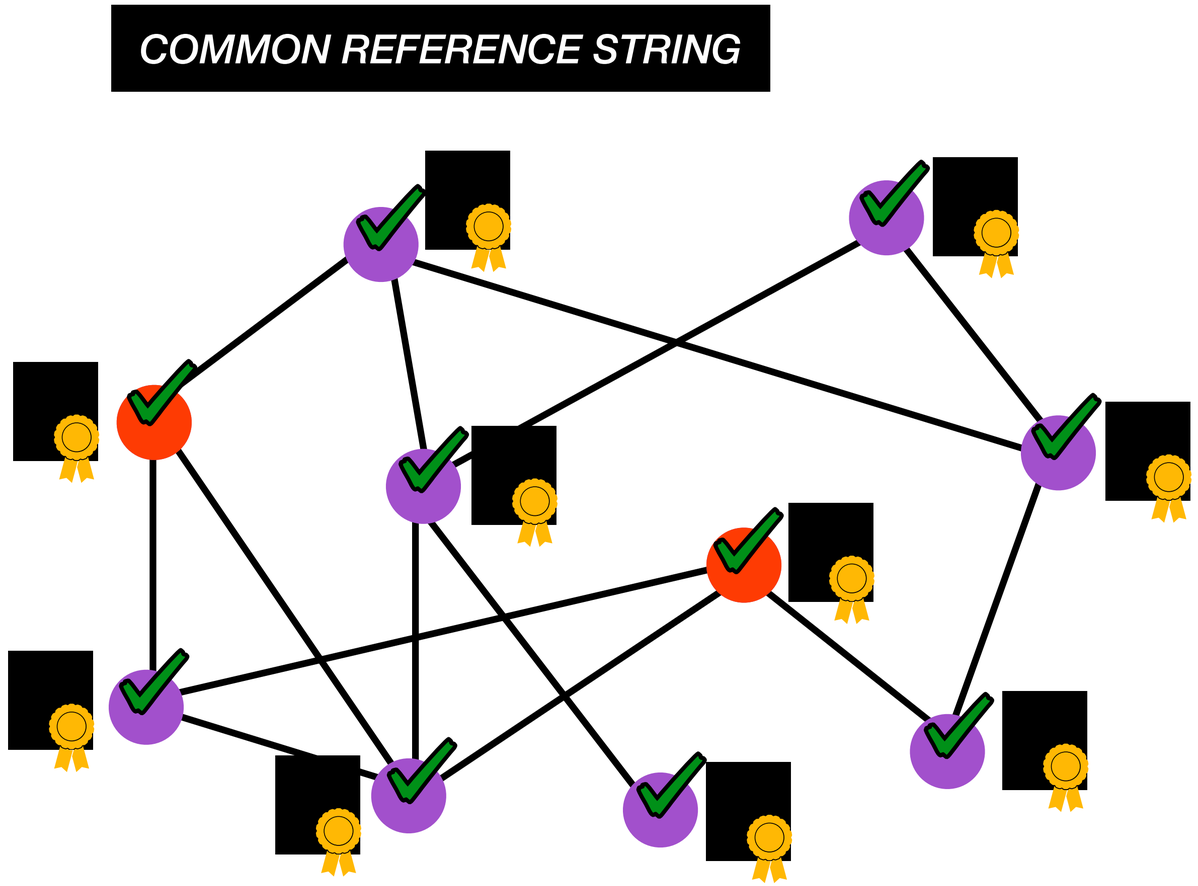
PROVER



[ABCMO23] constructs a distributed SNARG for any network property (in P)!

what if the adversary corrupts some nodes?

distributed SNARGs



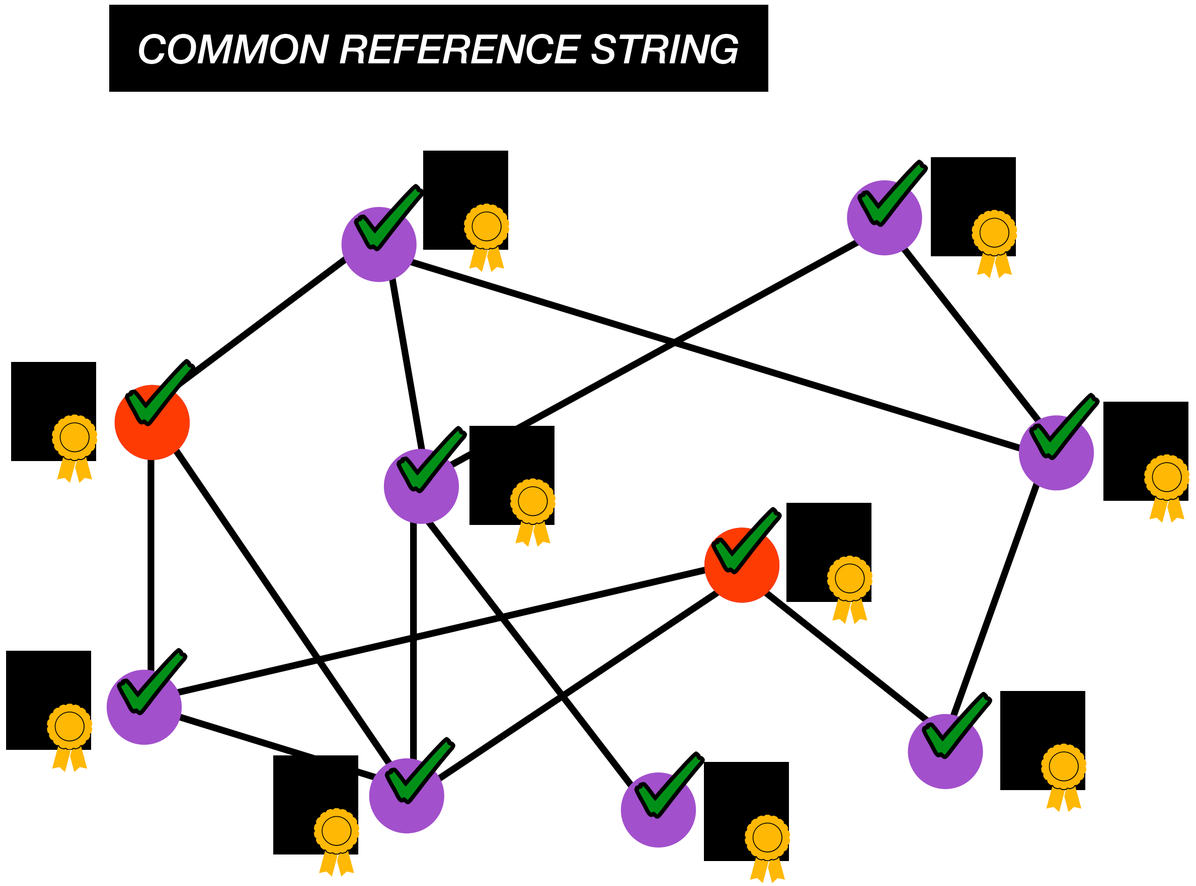
[ABCMO23] constructs a distributed SNARG for any network property (in P)!

what if the adversary corrupts some nodes?

no soundness guarantee :(

distributed SNARGs

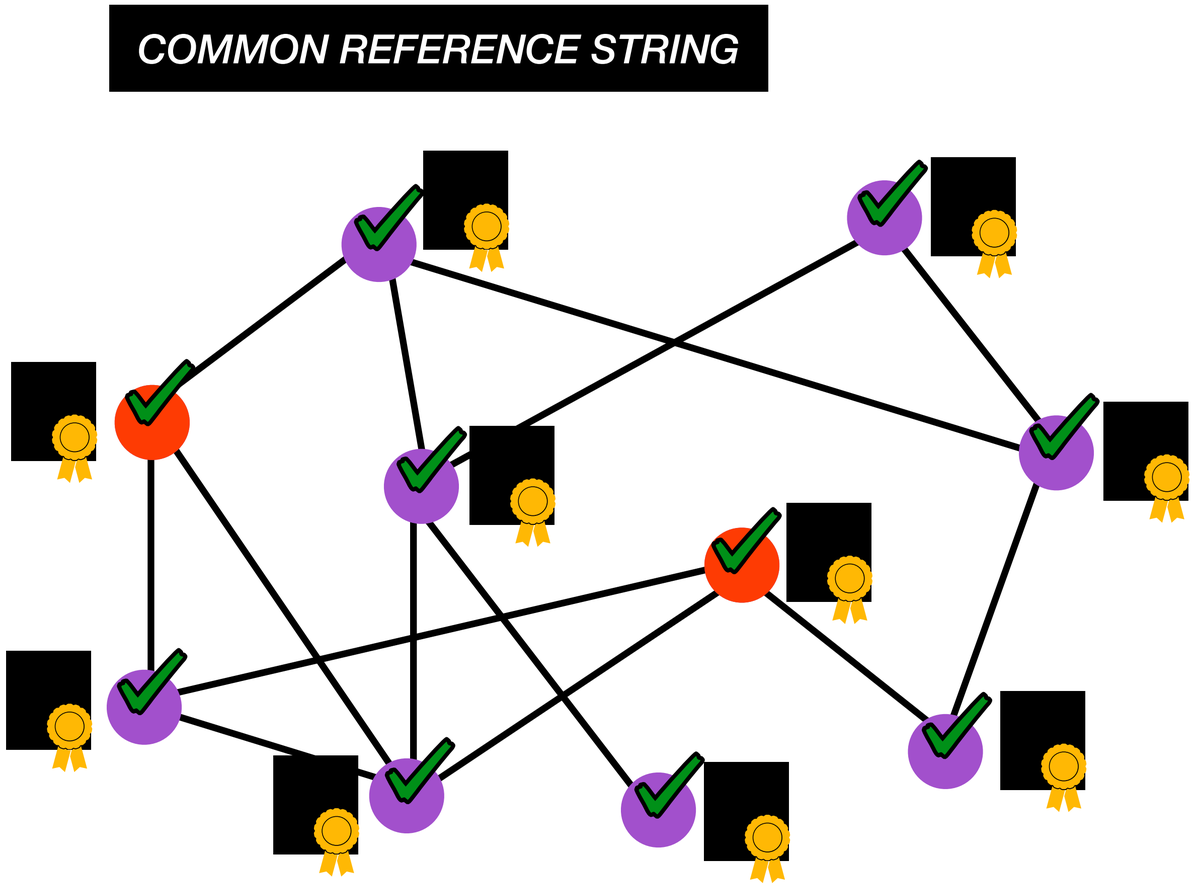
PROVER



what if the adversary corrupts some nodes?

distributed SNARGs

PROVER



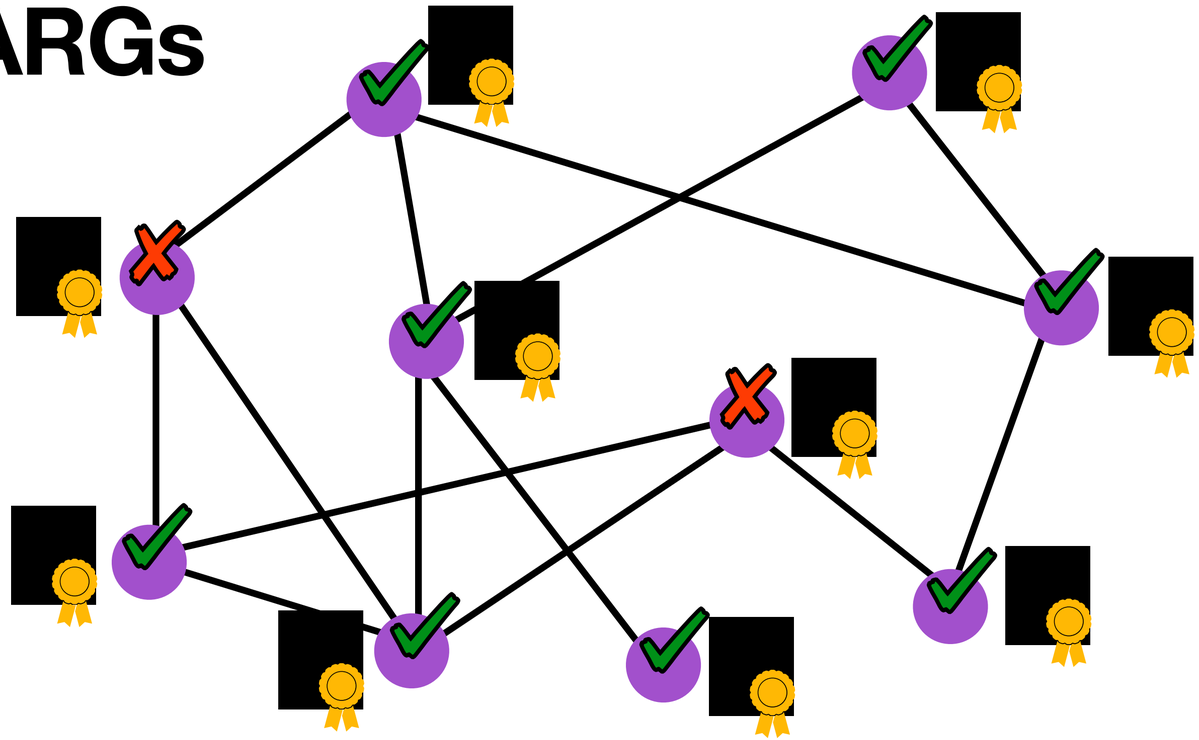
what if the adversary corrupts some nodes?

let's define the model

COMMON REFERENCE STRING

this work: distributed SNARGs on corrupted network

PROVER

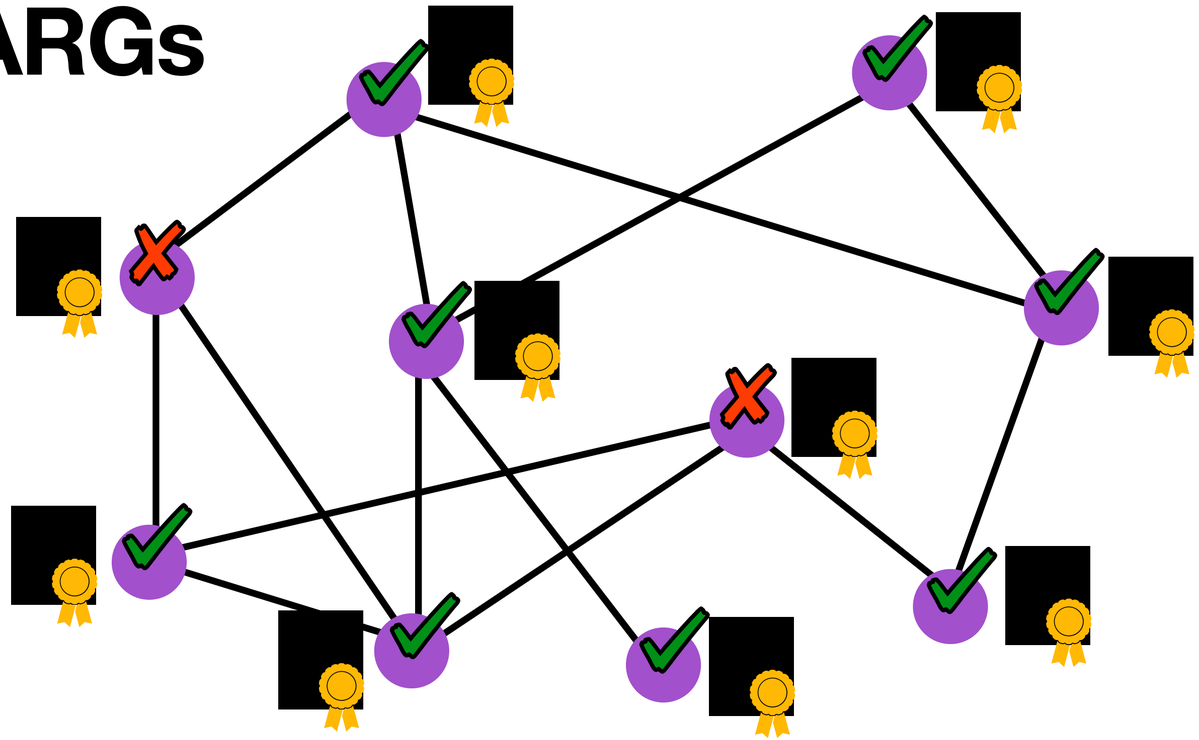


COMPLETENESS all verifiers accept honestly generated certificates

COMMON REFERENCE STRING

this work: distributed SNARGs on corrupted network

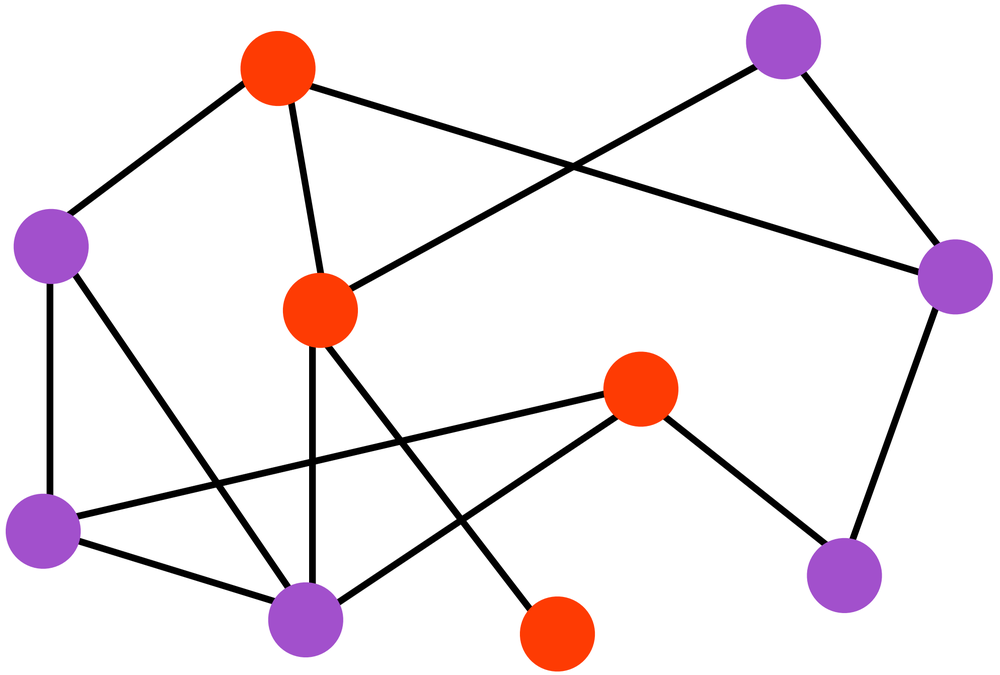
PROVER



COMPLETENESS all verifiers accept honestly generated certificates

what is the right notion of soundness to hope for?

network assumptions

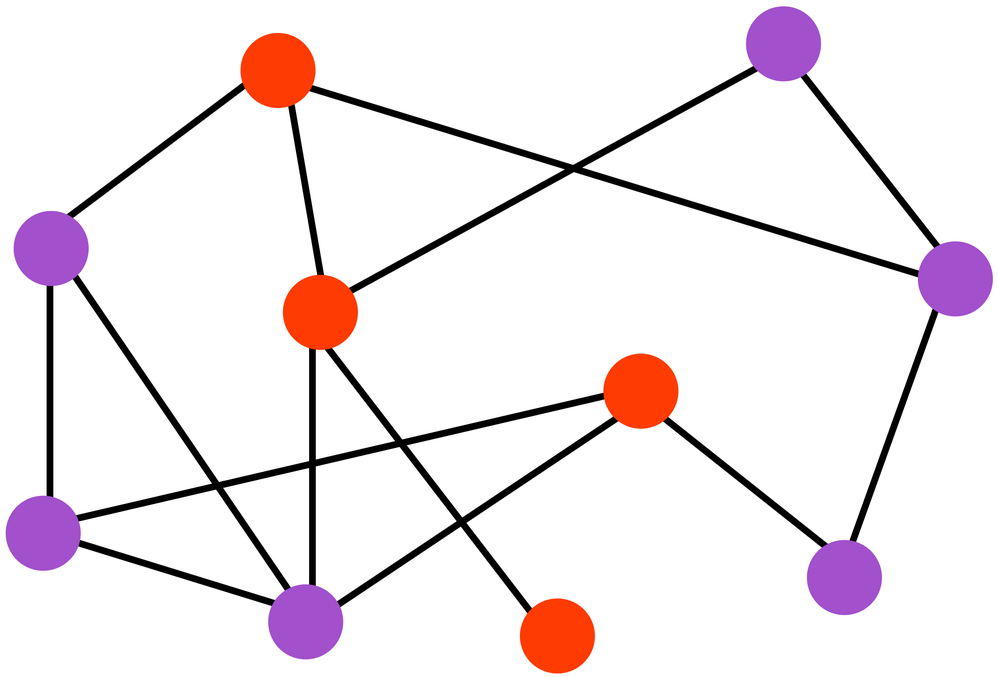


= honest node

= corrupt node

network assumptions

for our construction
we'll need to make assumptions:



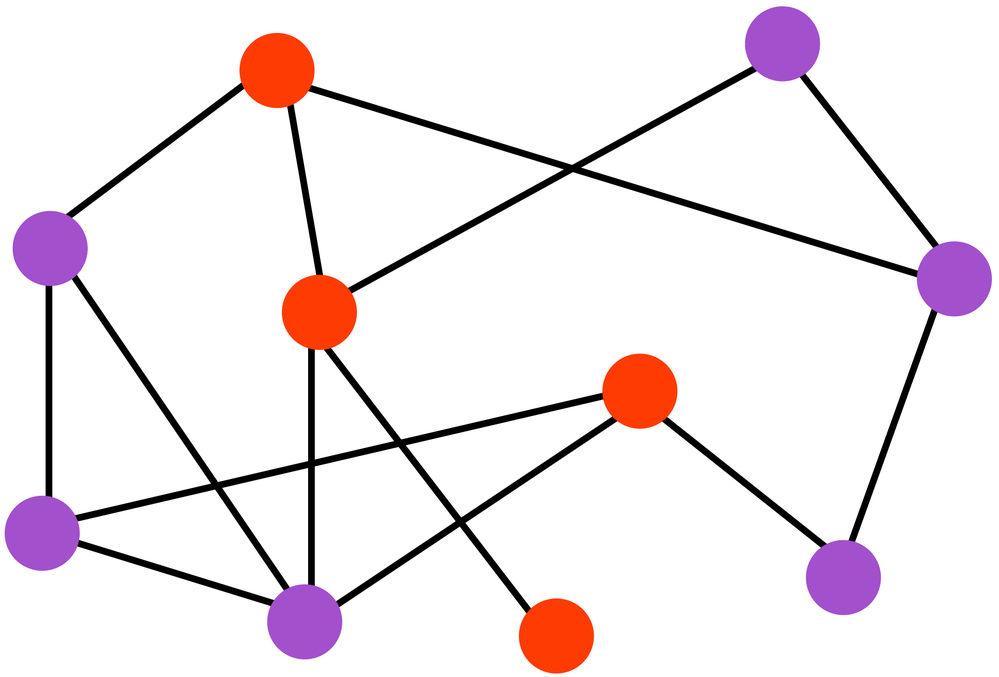
= honest node

= corrupt node

network assumptions

<u>for our construction</u>
<u>we'll need to make assumptions:</u>

- honest nodes remain connected

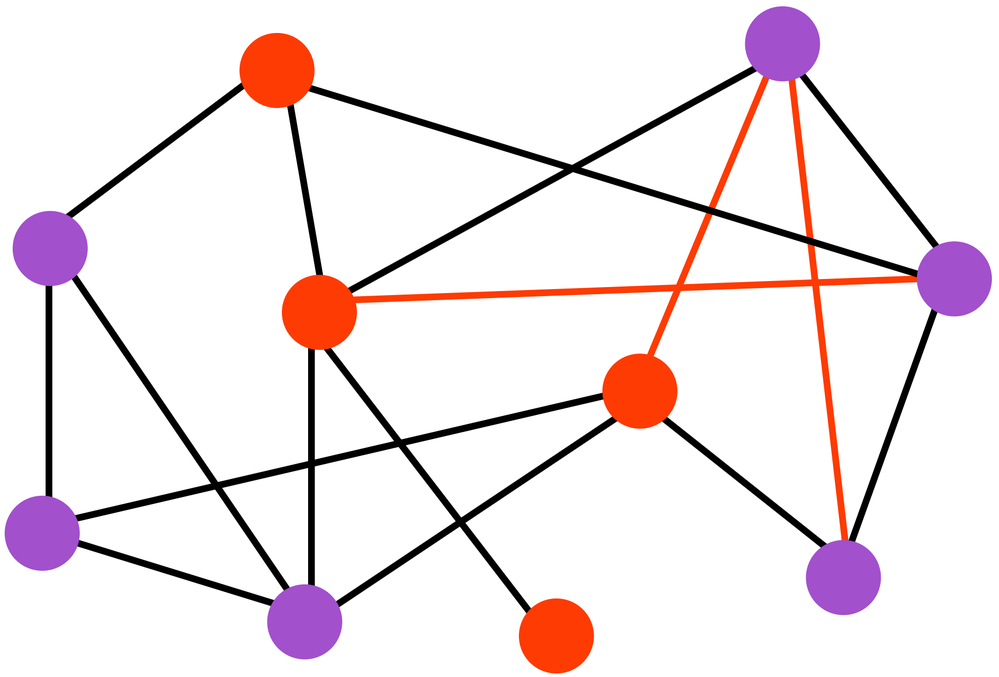


= honest node
= corrupt node

network assumptions

<u>for our construction</u>
<u>we'll need to make assumptions:</u>

- honest nodes remain connected

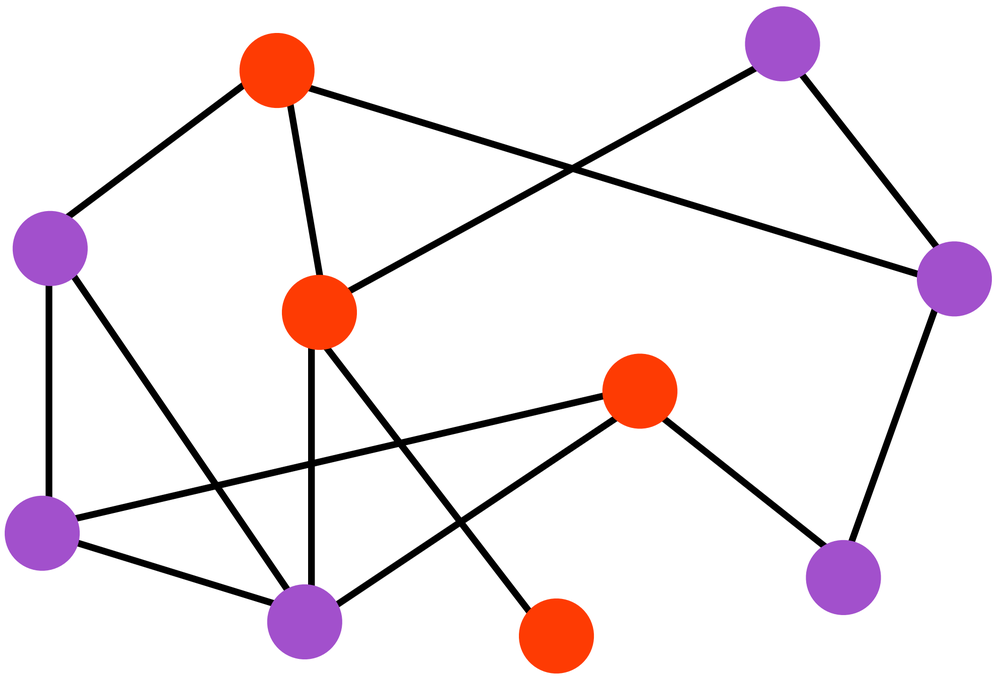


= honest node
= corrupt node

network assumptions

<u>for our construction</u>
<u>we'll need to make assumptions:</u>

- honest nodes remain connected

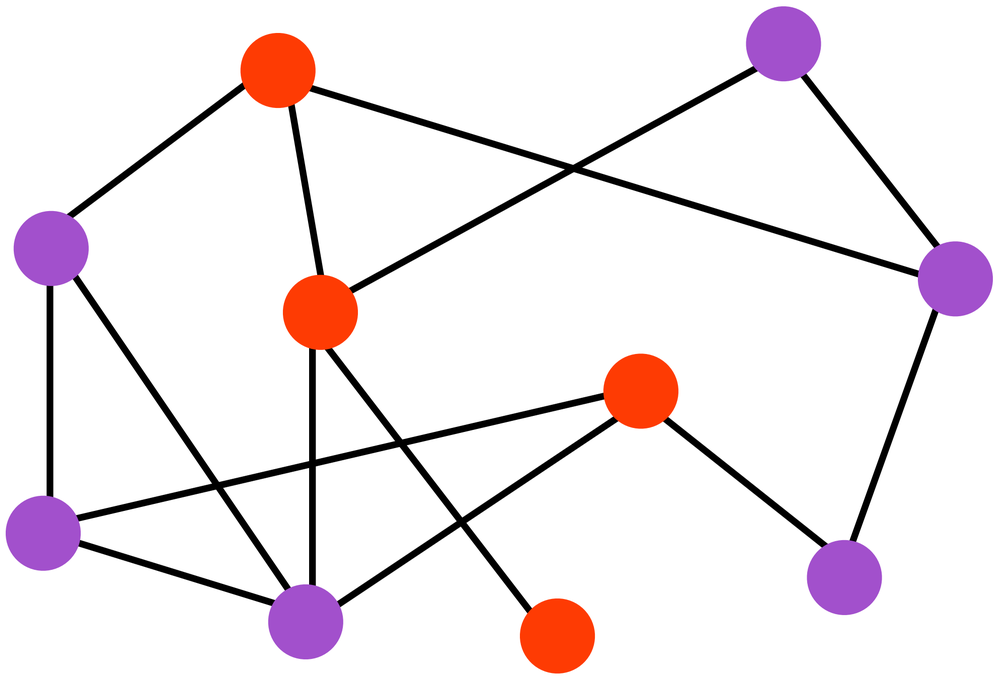


= honest node
= corrupt node

network assumptions

for our construction
we'll need to make assumptions:

- honest nodes remain connected
- every corrupt node has an honest neighbor



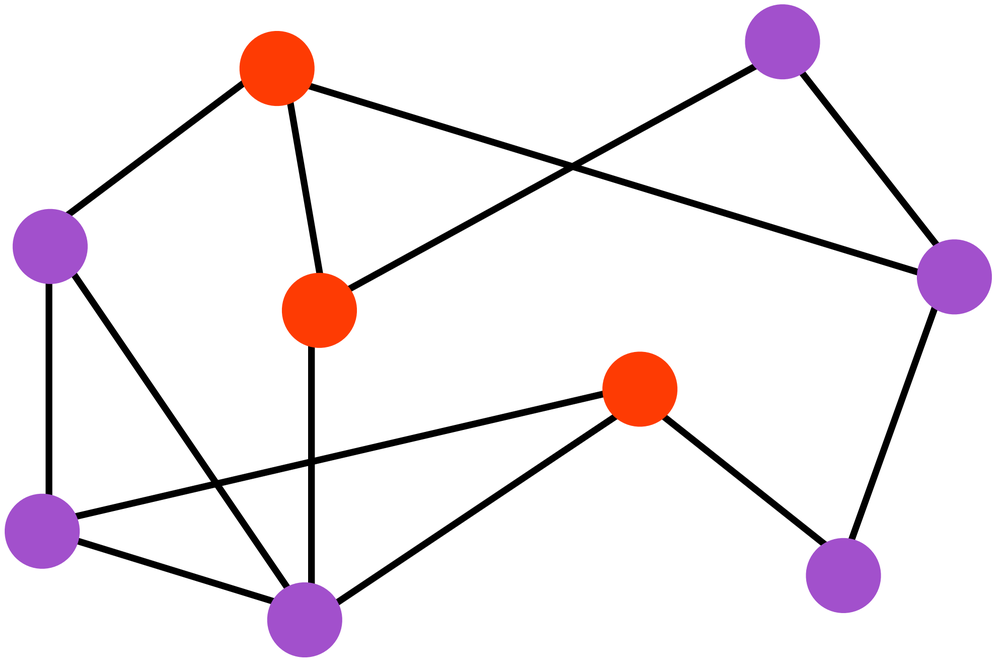
= honest node

= corrupt node

network assumptions

<u>for our construction</u>
<u>we'll need to make assumptions:</u>

- honest nodes remain connected
- every corrupt node has an honest neighbor

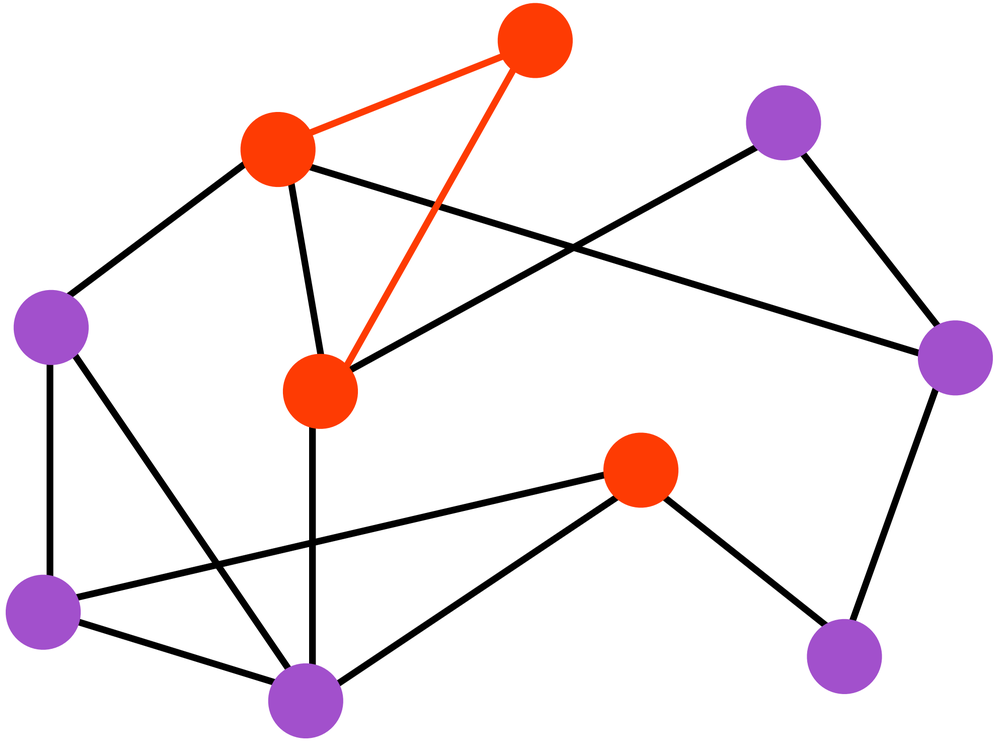


= honest node
= corrupt node

network assumptions

<u>for our construction</u>
<u>we'll need to make assumptions:</u>

- honest nodes remain connected
- every corrupt node has an honest neighbor

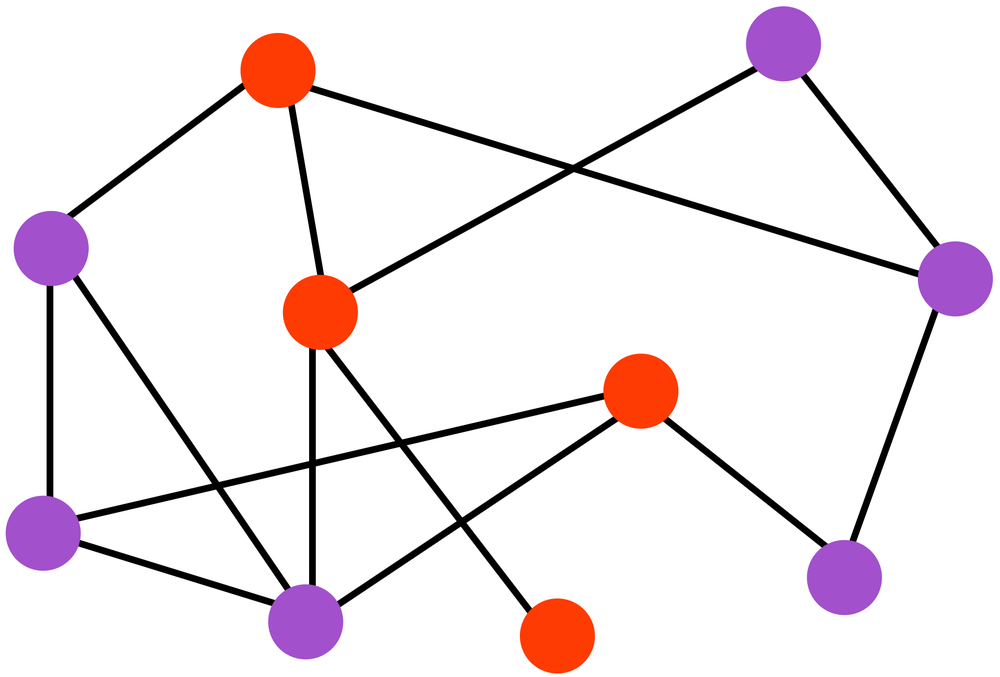


= honest node
= corrupt node

network assumptions

for our construction
we'll need to make assumptions:

- honest nodes remain connected
- every corrupt node has an honest neighbor



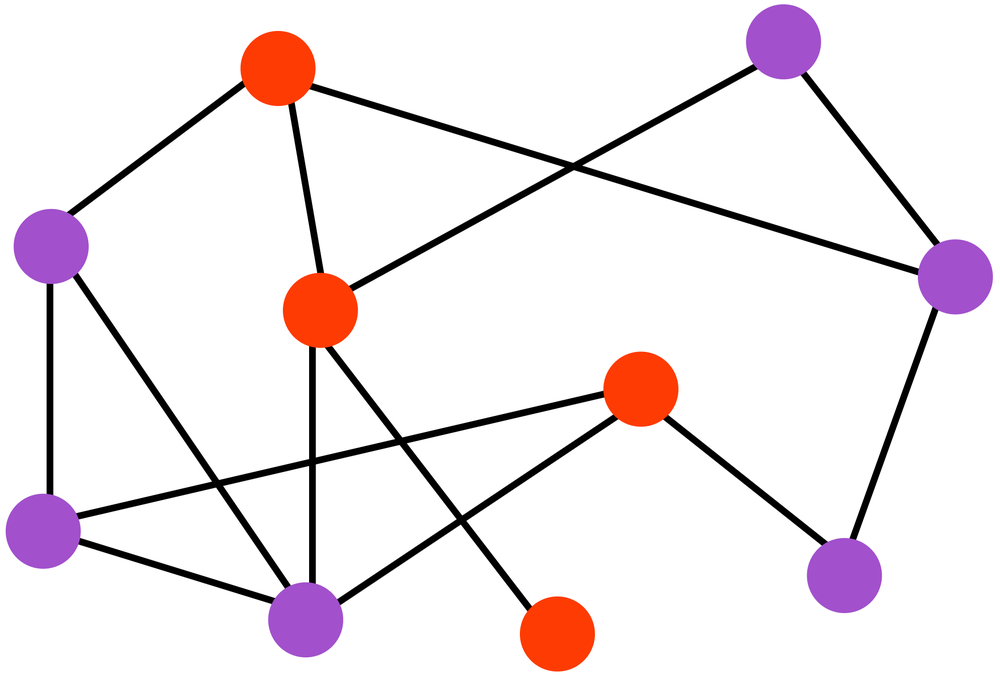
= honest node

= corrupt node

network assumptions

for our construction
we'll need to make assumptions:

- honest nodes remain connected
- every corrupt node has an honest neighbor
- nodes know the network size



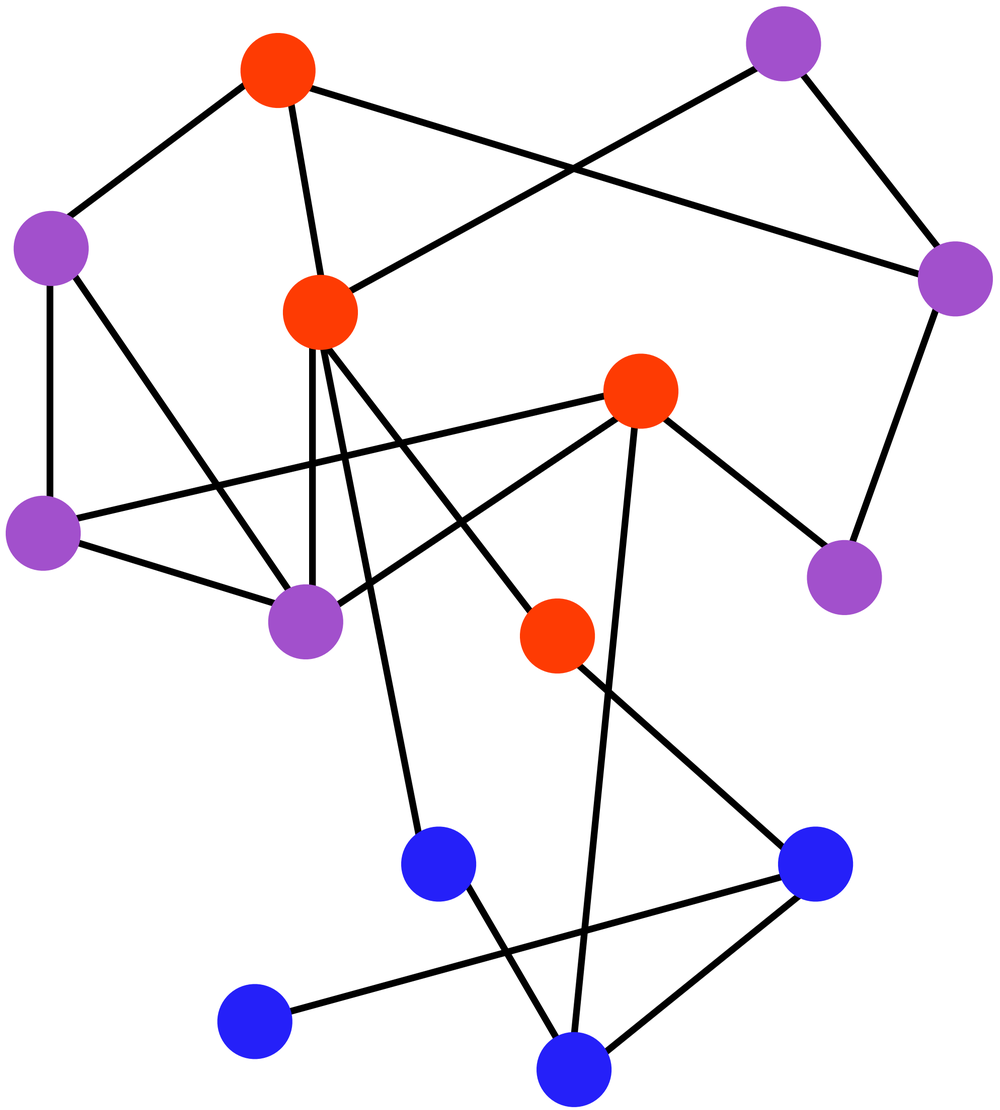
= honest node
= corrupt node

network assumptions

for our construction
we'll need to make assumptions:

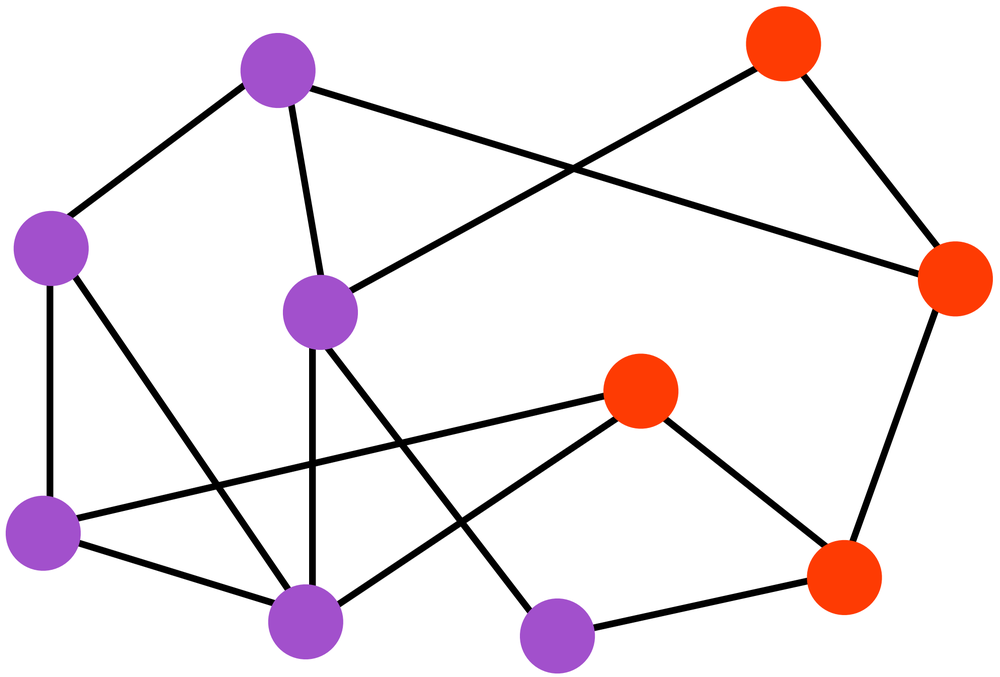
- honest nodes remain connected
- every corrupt node has an honest neighbor
- nodes know the network size

= honest node
= corrupt node
= "ghost" node



soundness notion

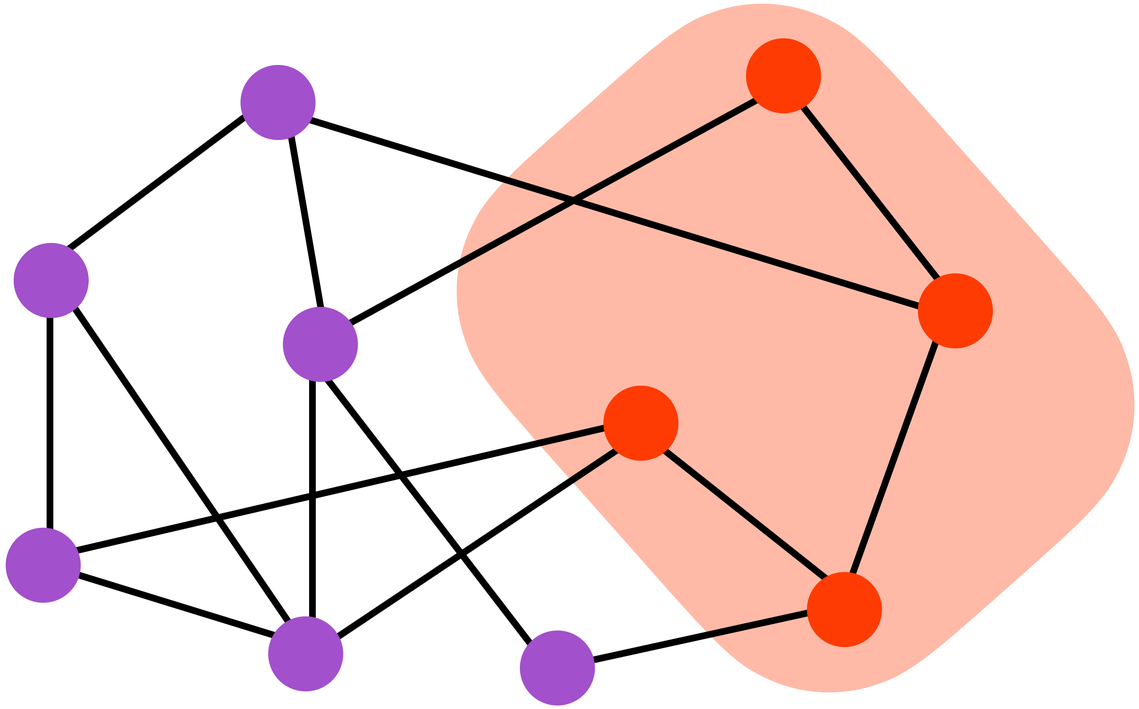
honest nodes cannot verify anything about edges between corrupt nodes



SOUNDNESS if the network **doesn't satisfy** the property, no efficient cheating prover can trick all verifiers into accepting

soundness notion

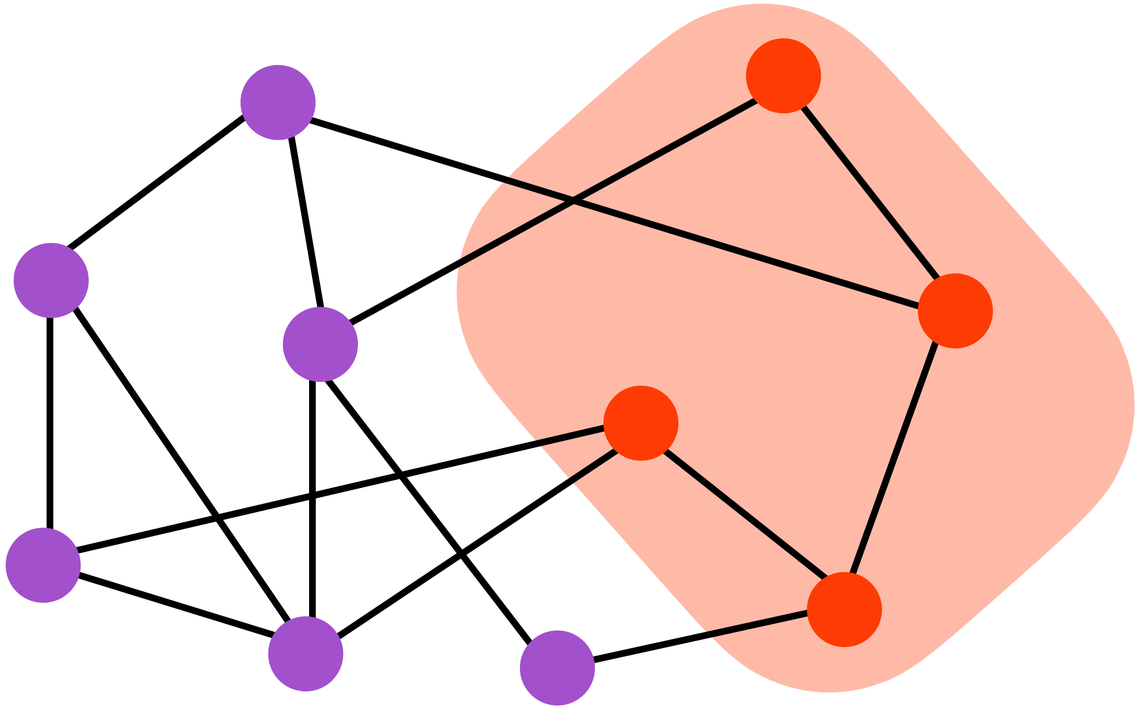
honest nodes cannot verify anything about edges between corrupt nodes



SOUNDNESS if the network **doesn't satisfy** the property, no efficient cheating prover can trick all verifiers into accepting

soundness notion

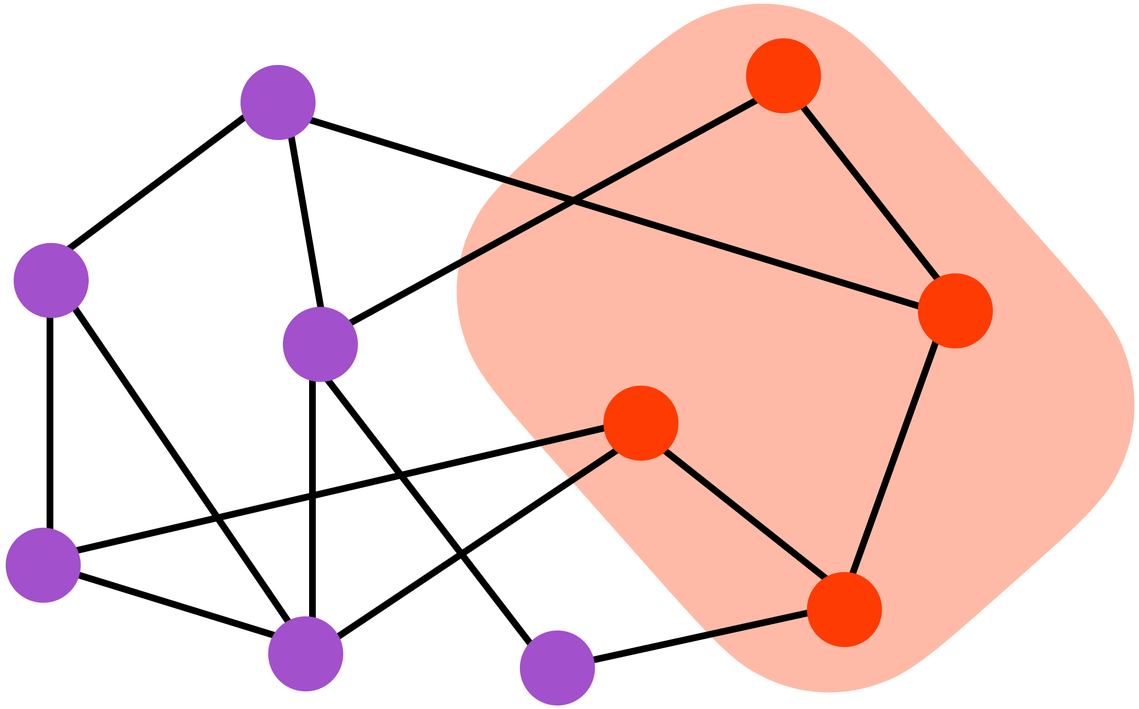
honest nodes cannot verify anything about edges between corrupt nodes



SOUNDNESS if the network **doesn't satisfy** the property, no efficient cheating prover can trick all verifiers into accepting

soundness notion

honest nodes cannot verify anything about edges between corrupt nodes

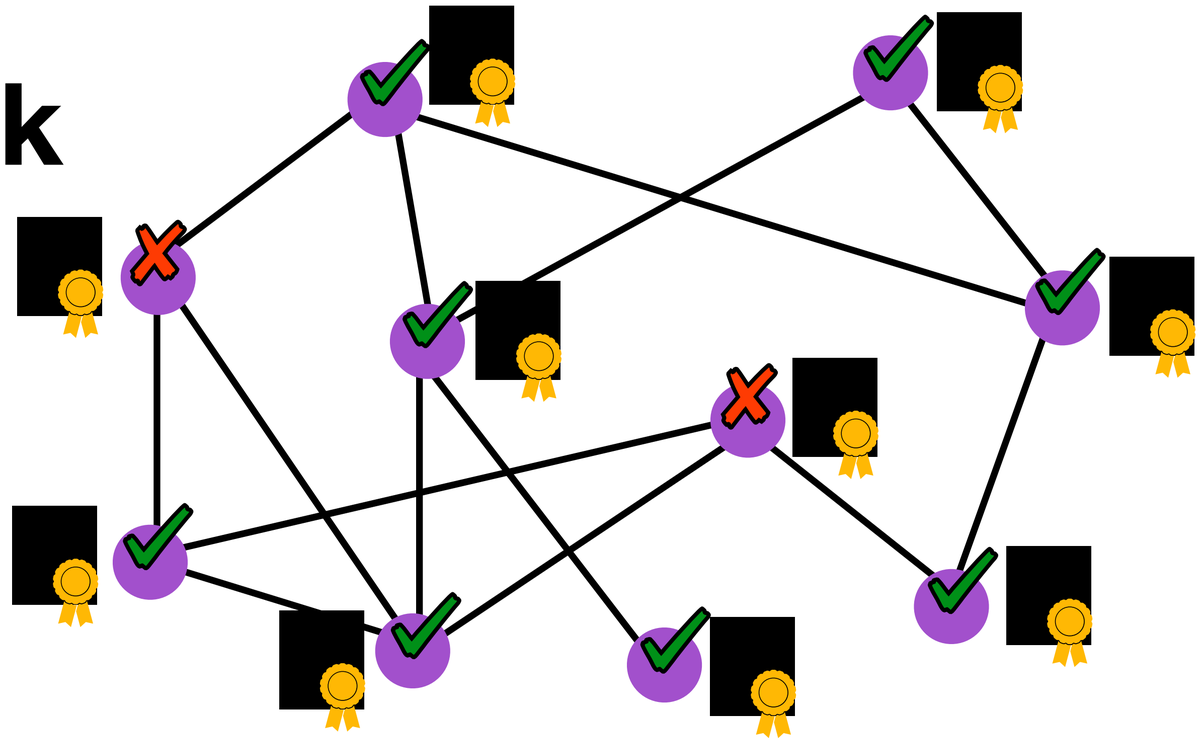


SOUNDNESS if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

~~***SOUNDNESS*** if the network **doesn't satisfy** the property, no efficient cheating prover can trick all verifiers into accepting~~

distributed SNARGs on corrupted network

PROVER



COMPLETENESS all verifiers accept honestly generated certificates

SOUNDNESS if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

distributed SNARGs on corrupted network

n = network size, t = # of corrupt nodes

distributed SNARGs on corrupted network

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

distributed SNARGs on corrupted network

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

by using fully local SEH families and RAM SNARGs

distributed SNARGs on corrupted network

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

by using fully local SEH families and RAM SNARGs

yay! :)

distributed SNARGs on corrupted network

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

by using fully local SEH families and RAM SNARGs

yay! :)

could we hope for shorter certificates?

succinctness lower bound

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

succinctness lower bound

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

distributed SNARG for corrupted networks *more succinct* than ours $\Longrightarrow$ **SNARG for NP**

succinctness lower bound

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

distributed SNARG for corrupted networks *more succinct* than ours $\Longrightarrow$ **SNARG for NP**

not currently known from the tools we're using!

succinctness lower bound

n = network size, t = # of corrupt nodes

we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

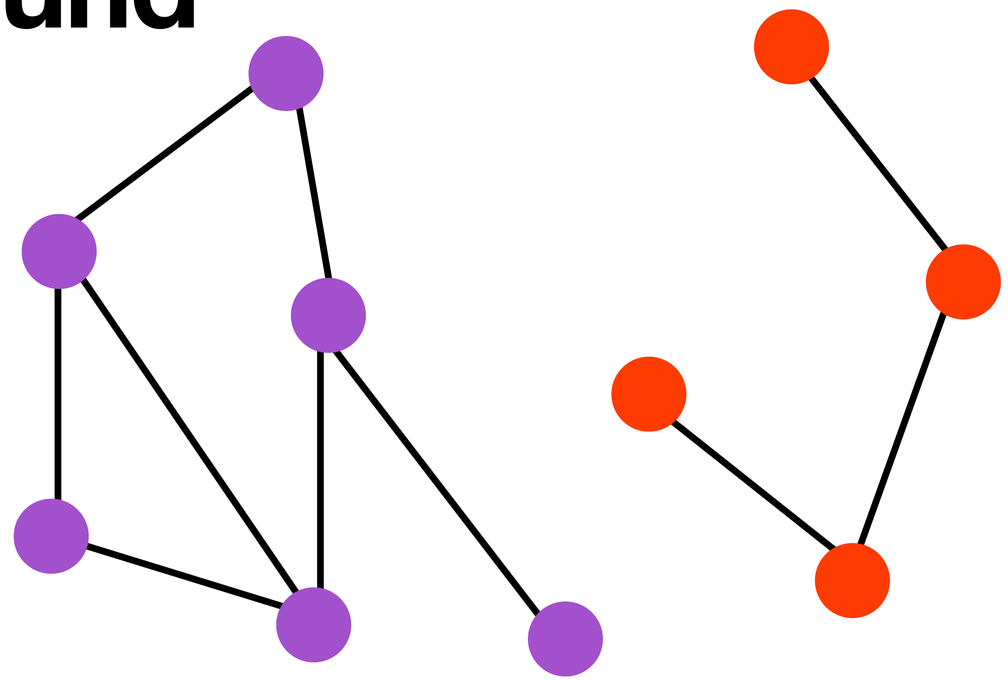
distributed SNARG for corrupted networks *more succinct* than ours $\Longrightarrow$ **SNARG for NP**

not currently known from the tools we're using!

inspired by lower bounds showing arguments of proximity for P which are *succinct enough* imply a SNARG for NP [KR15,CJW25]

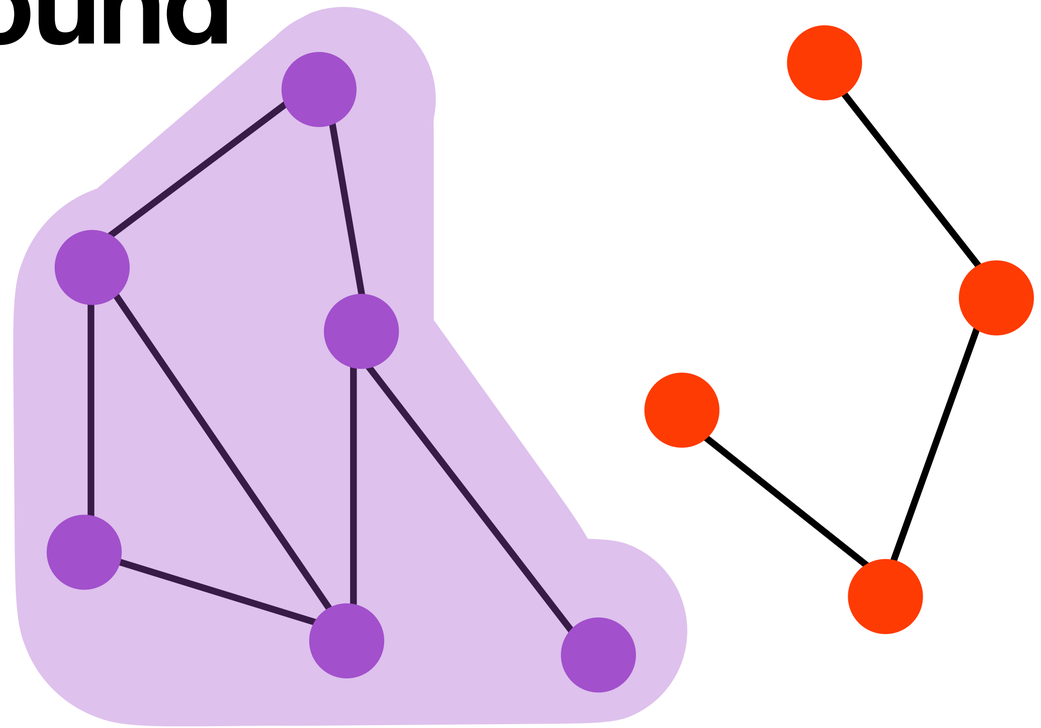
succinctness lower bound

let's try to encode the NP language we want a SNARG for into a corrupted network



succinctness lower bound

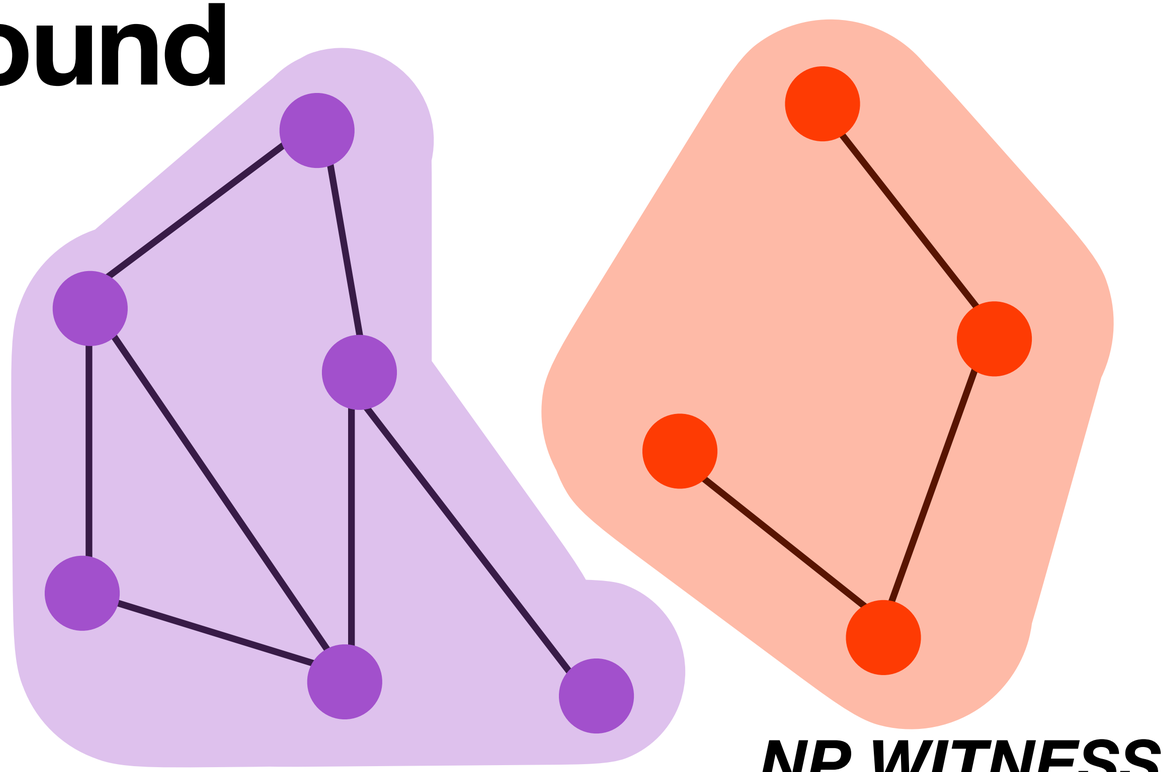
let's try to encode the NP language we want a SNARG for into a corrupted network



NP STATEMENT

succinctness lower bound

let's try to encode the NP language we want a SNARG for into a corrupted network

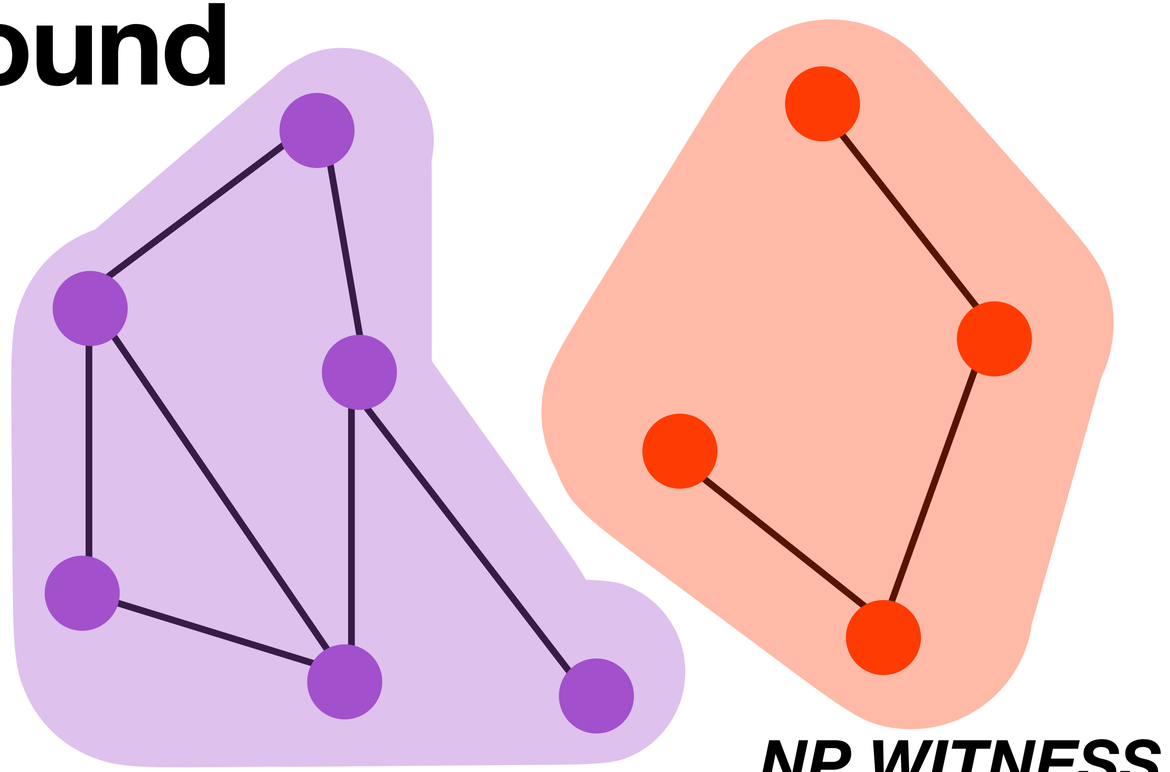


NP STATEMENT

NP WITNESS

succinctness lower bound

let's try to encode the NP language we want a SNARG for into a corrupted network



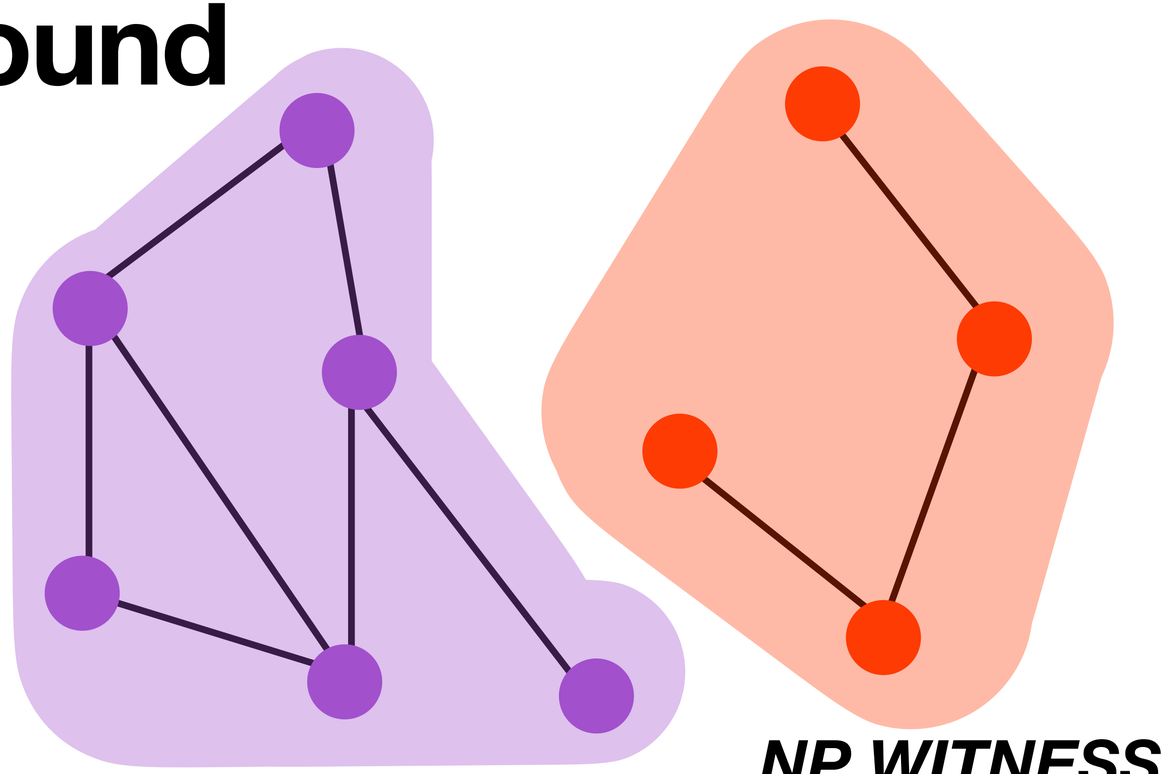
NP STATEMENT

NP WITNESS

property: network encodes a valid statement-witness pair

succinctness lower bound

let's try to encode the NP language we want a SNARG for into a corrupted network



NP STATEMENT

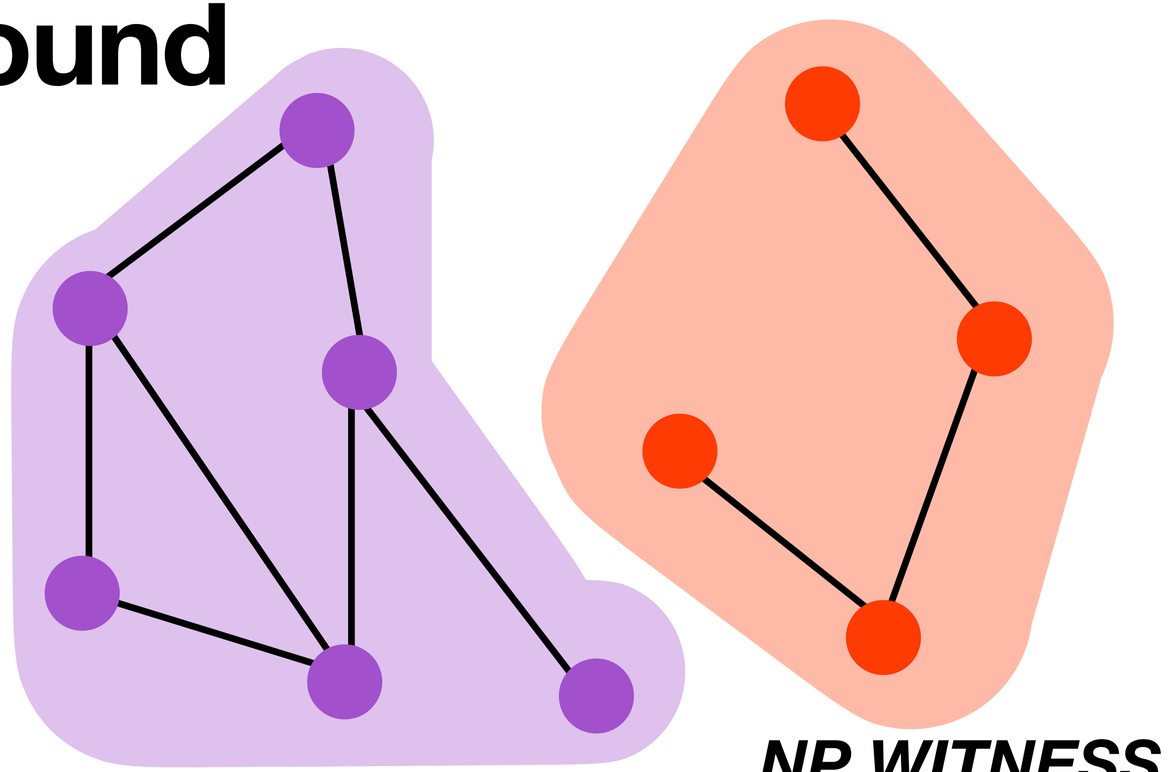
NP WITNESS

property: network encodes a valid statement-witness pair

but what about our network assumptions?

succinctness lower bound

let's try to encode the NP language we want a SNARG for into a corrupted network



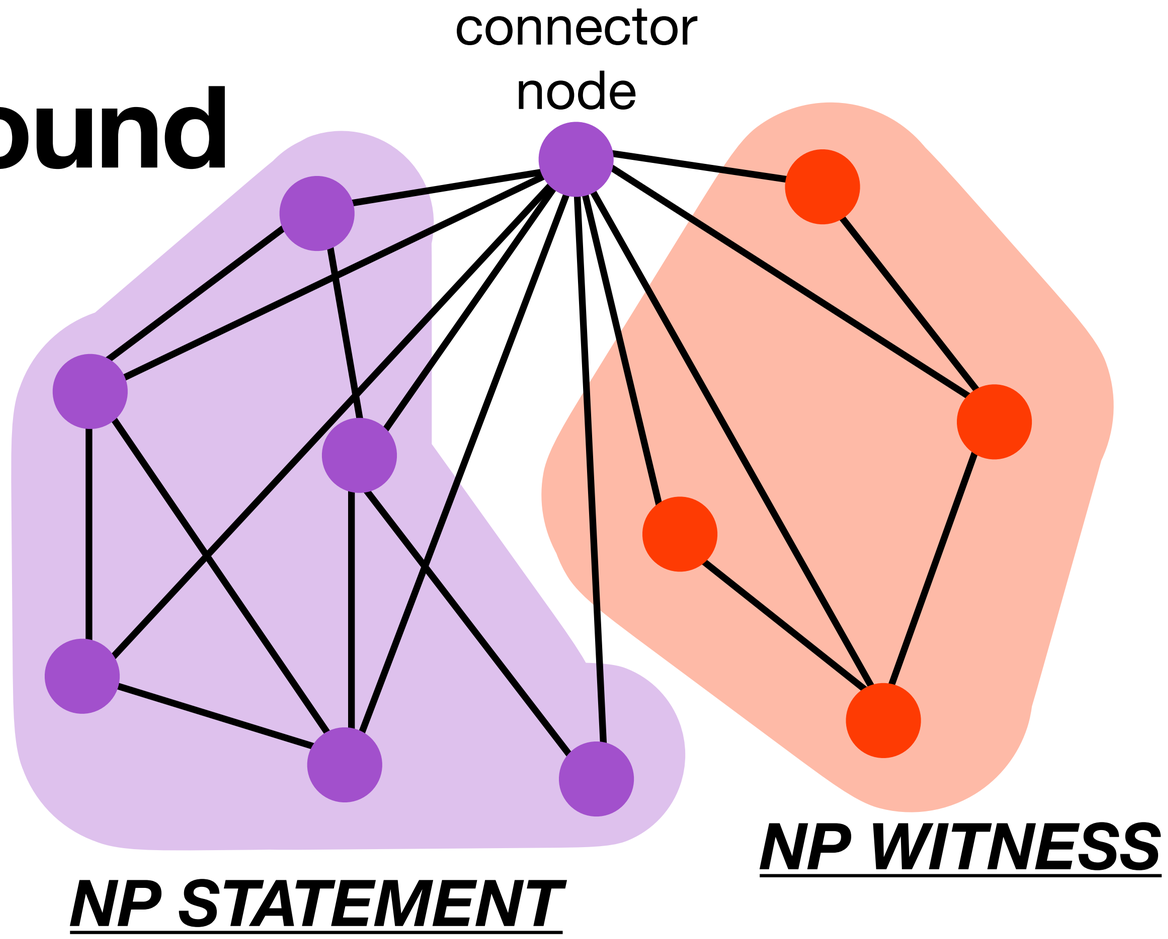
NP STATEMENT

NP WITNESS

property: network encodes a valid statement-witness pair

succinctness lower bound

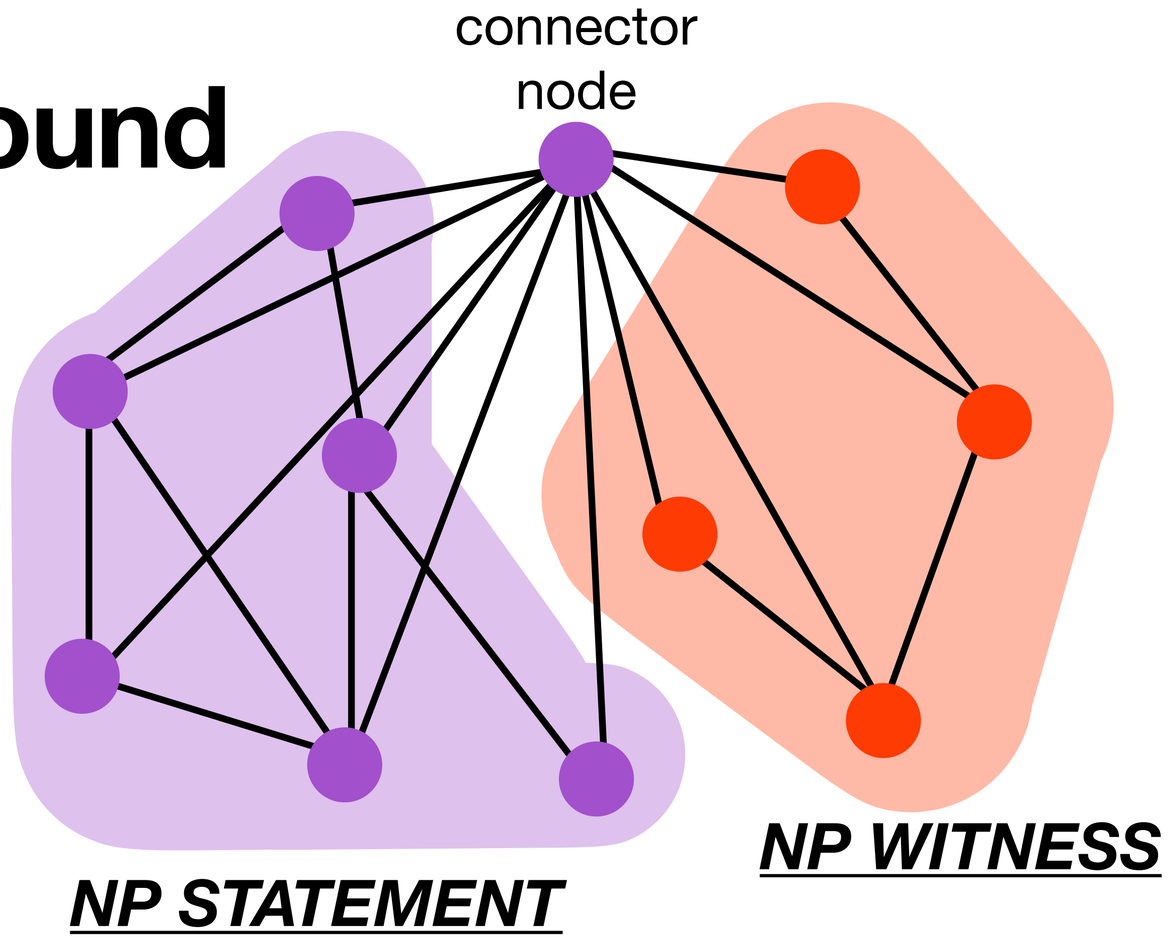
let's try to encode the NP language we want a SNARG for into a corrupted network



property: network encodes a valid statement-witness pair

succinctness lower bound

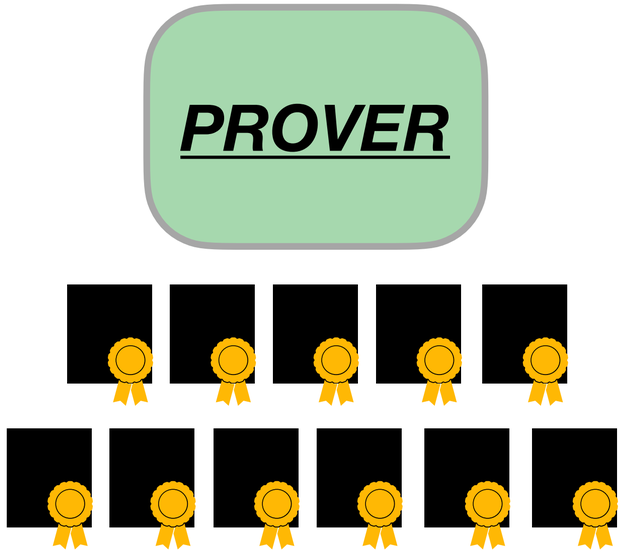
let's try to encode the NP language we want a SNARG for into a corrupted network

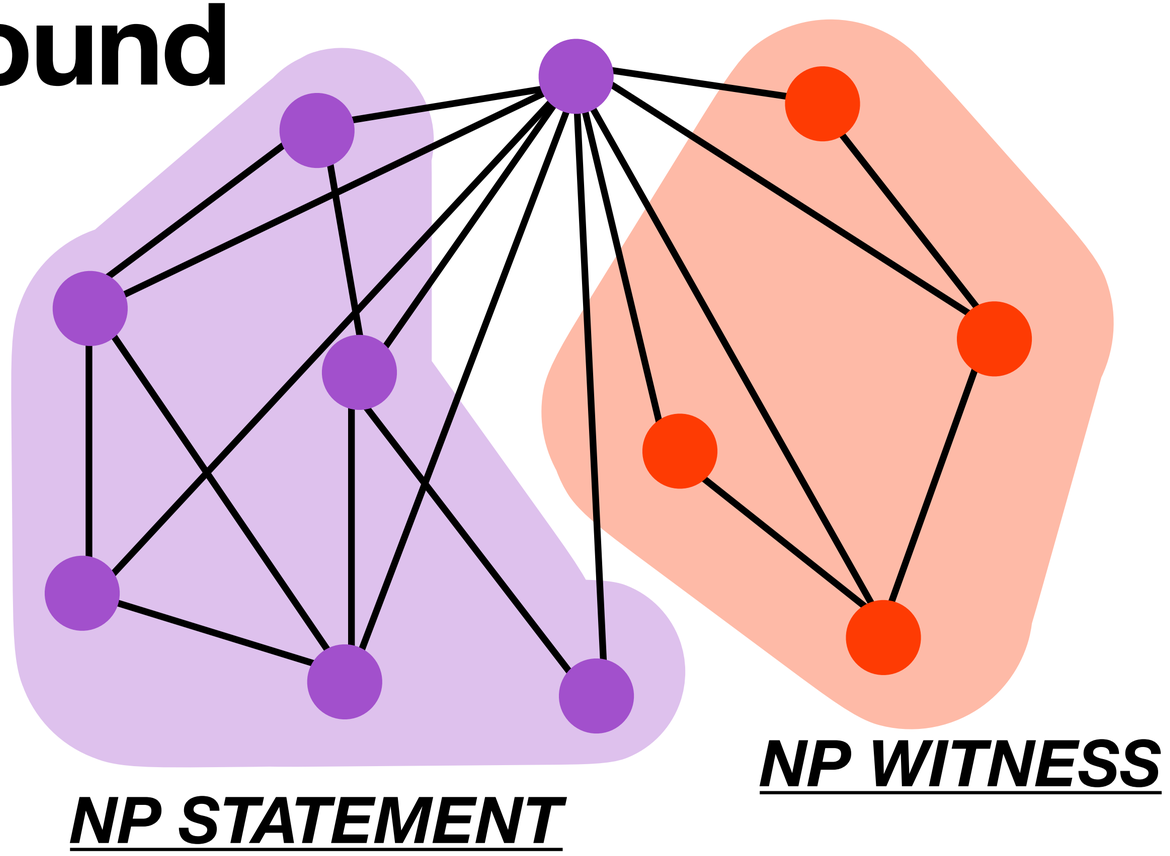


property: network encodes a valid statement-witness pair

we can use a distributed SNARG on this network/corruptions!

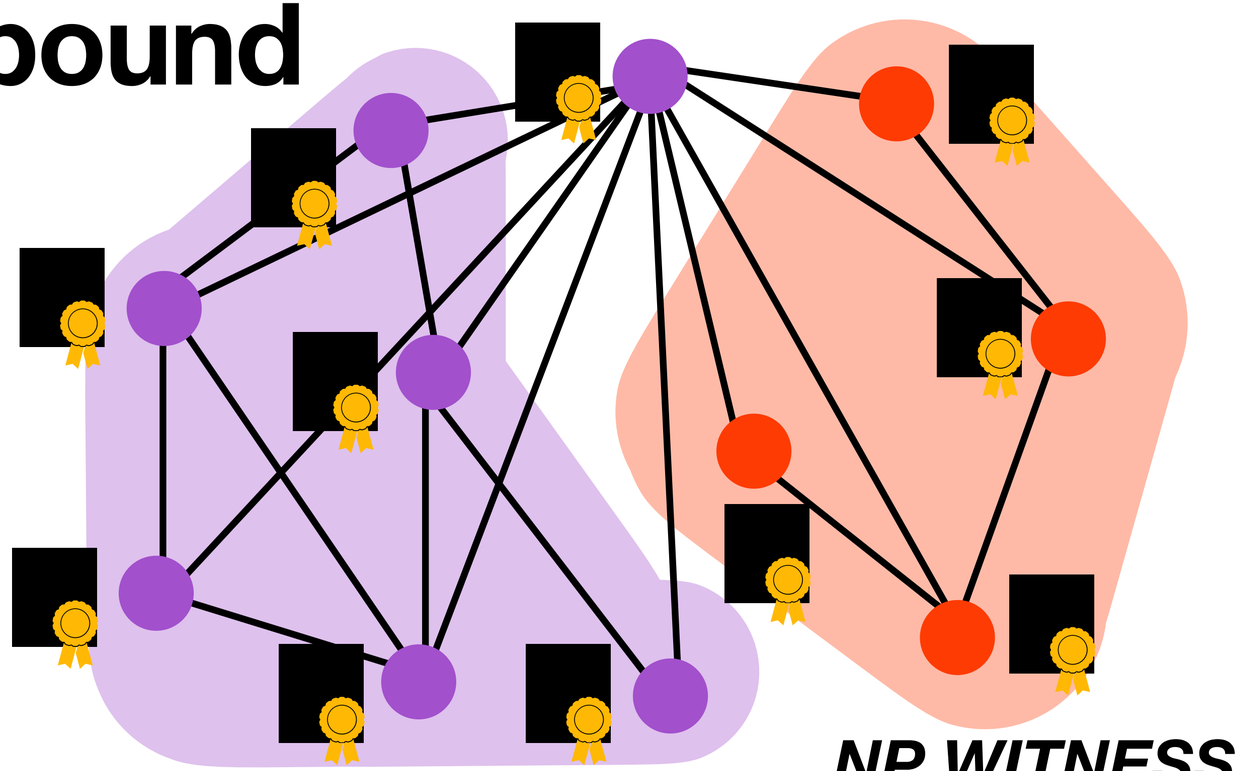
succinctness lower bound





succinctness lower bound

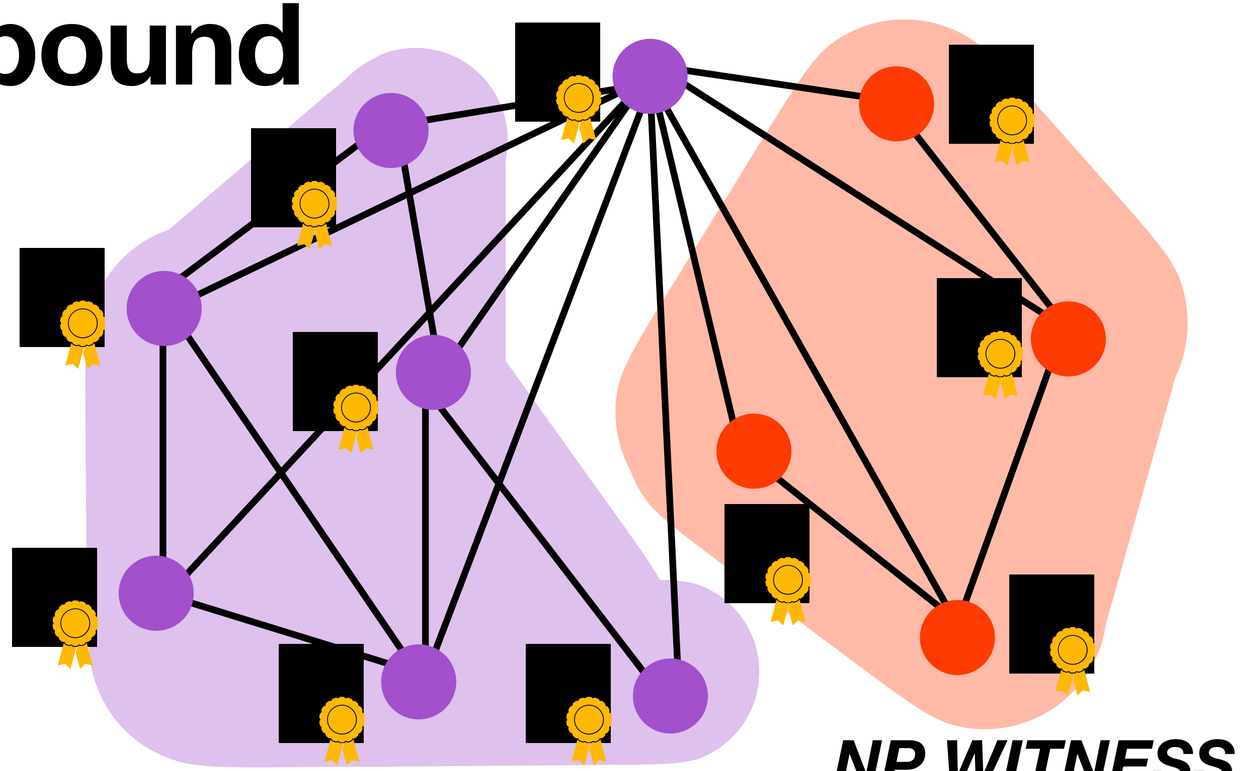
PROVER



NP STATEMENT

NP WITNESS

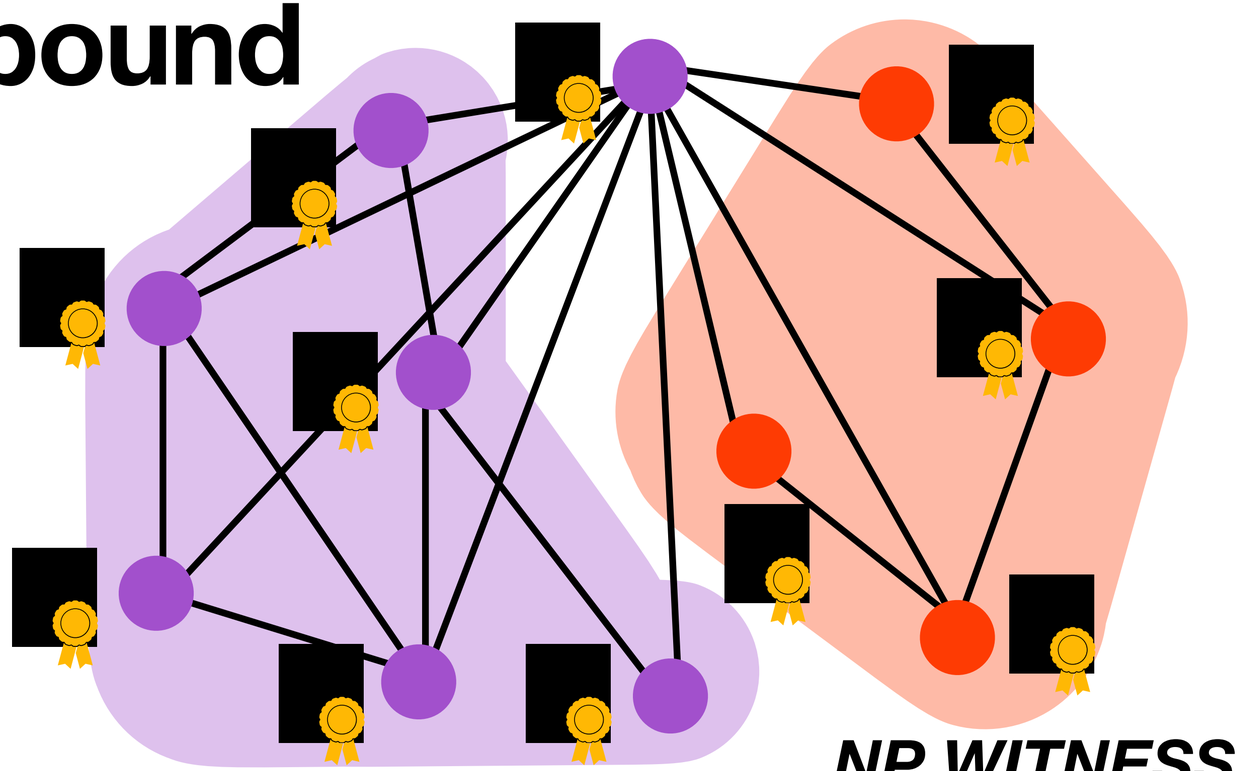
succinctness lower bound



if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

succinctness lower bound

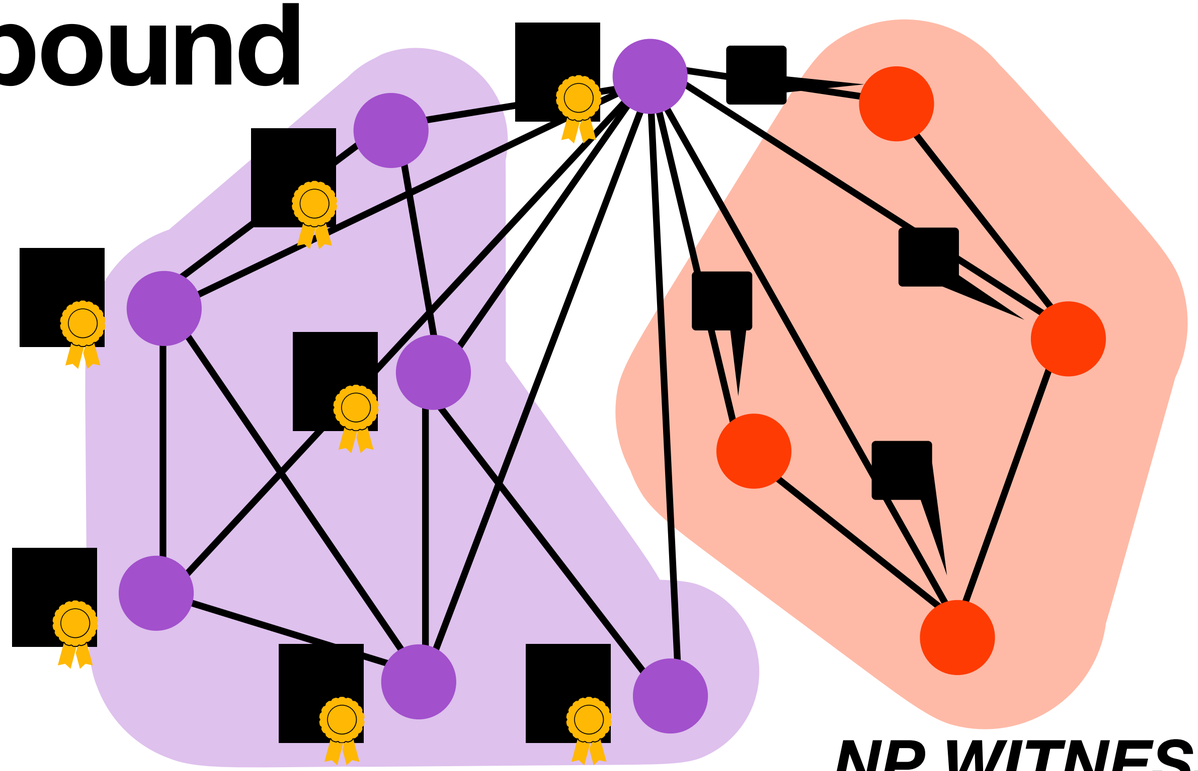
we only care about what the honest verifiers do... what do the honest verifiers need to verify?



if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

succinctness lower bound

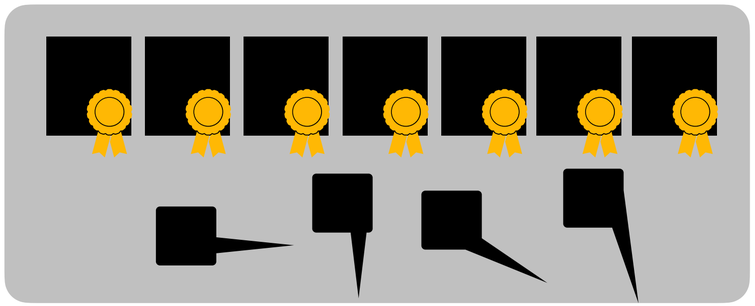
we only care about what the honest verifiers do... what do the honest verifiers need to verify?

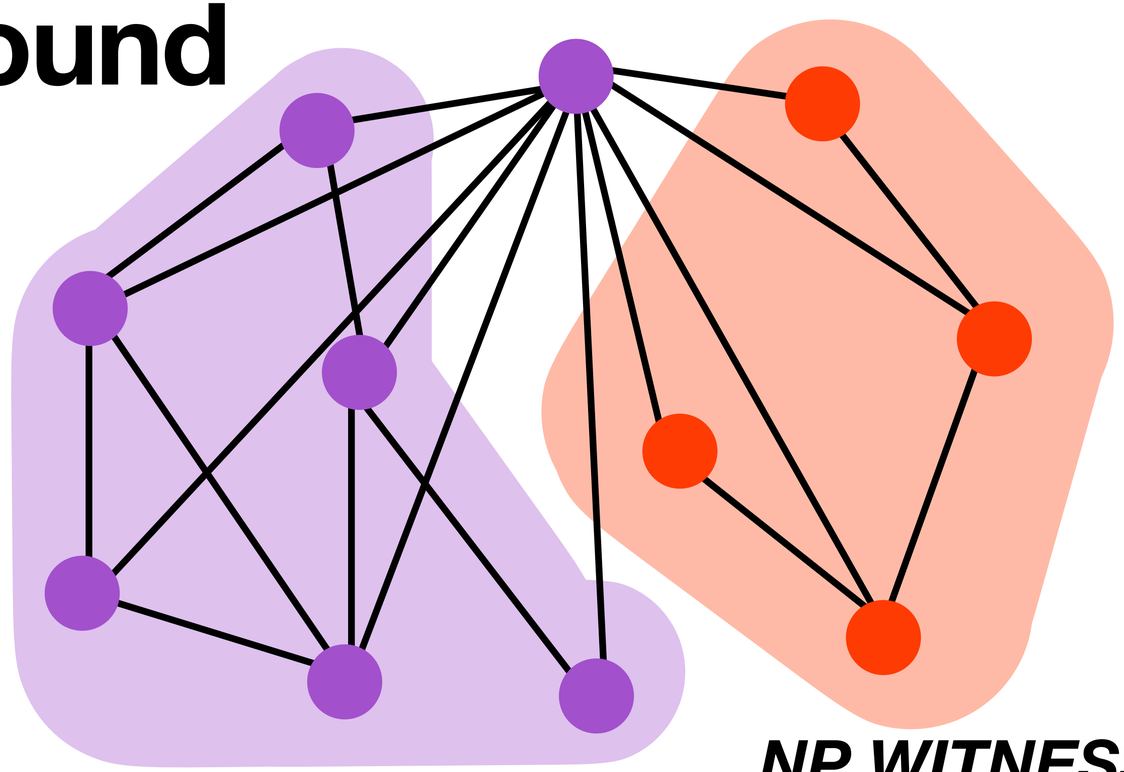


if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

succinctness lower bound

CERTIFICATE





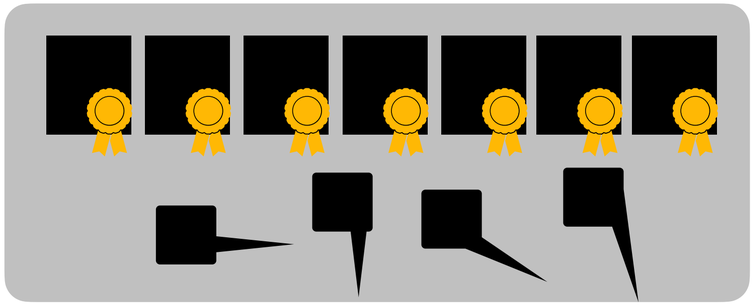
NP WITNESS

NP STATEMENT

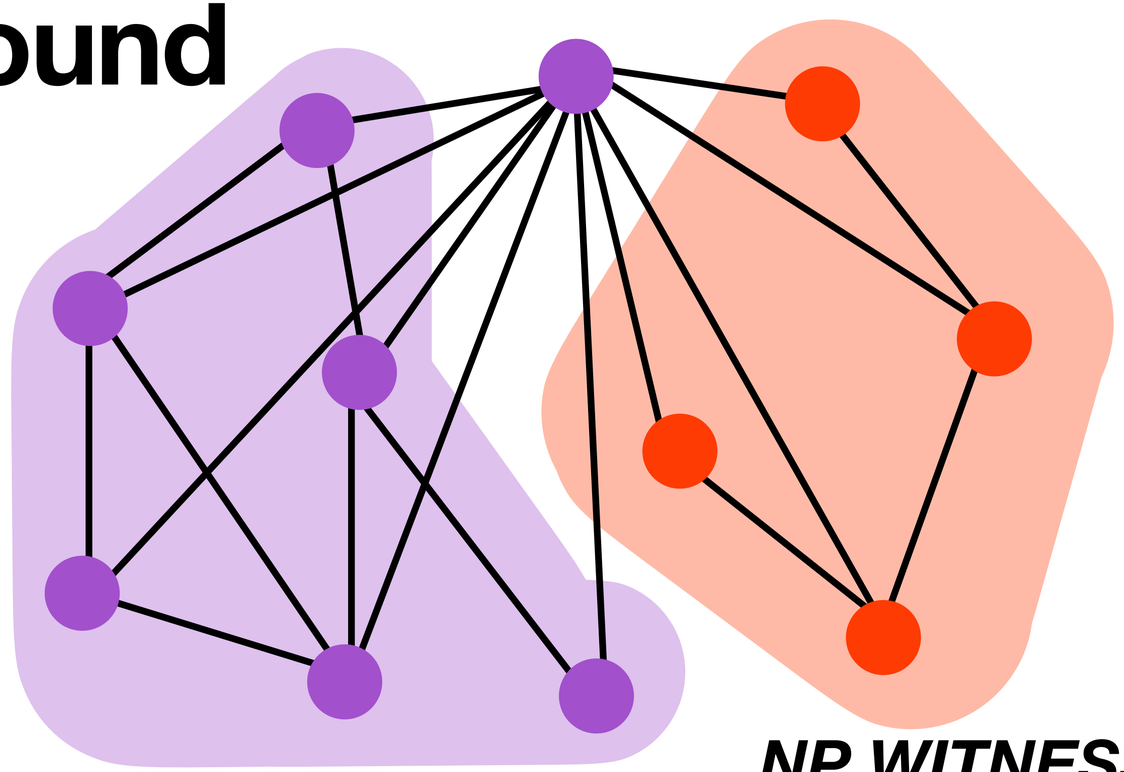
if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

succinctness lower bound

CERTIFICATE



if this is asymptotically smaller than the witness, it is a SNARG for NP!



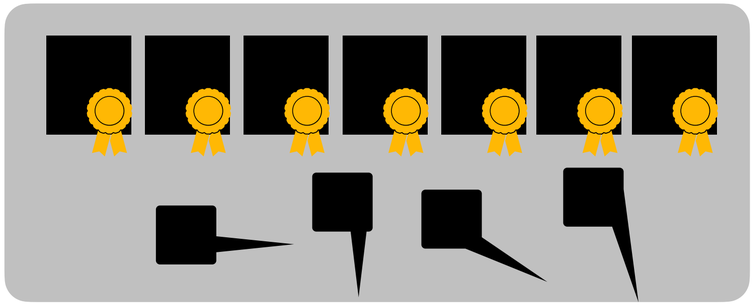
NP STATEMENT

NP WITNESS

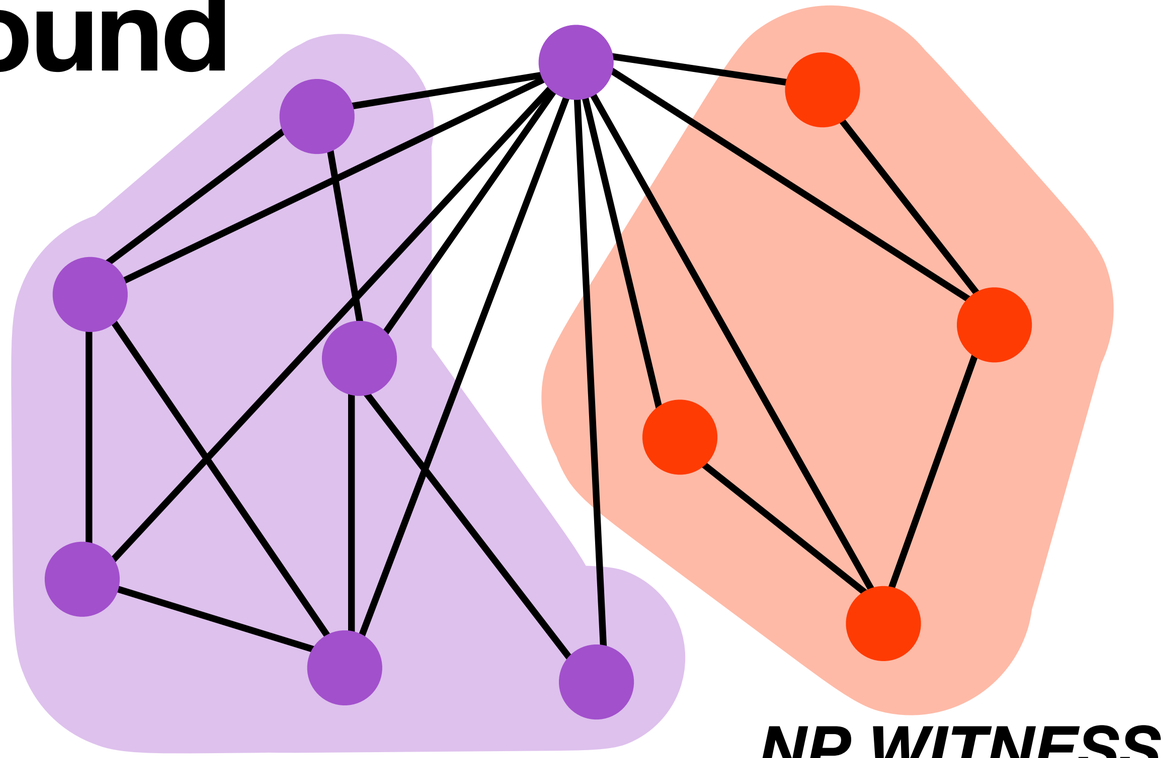
if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

succinctness lower bound

CERTIFICATE



if this is asymptotically smaller than the witness, it is a SNARG for NP!

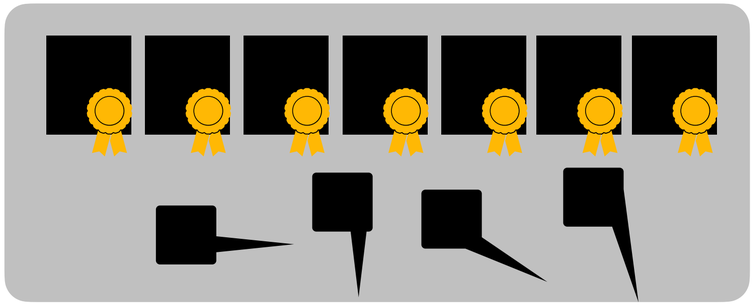


NP STATEMENT

NP WITNESS

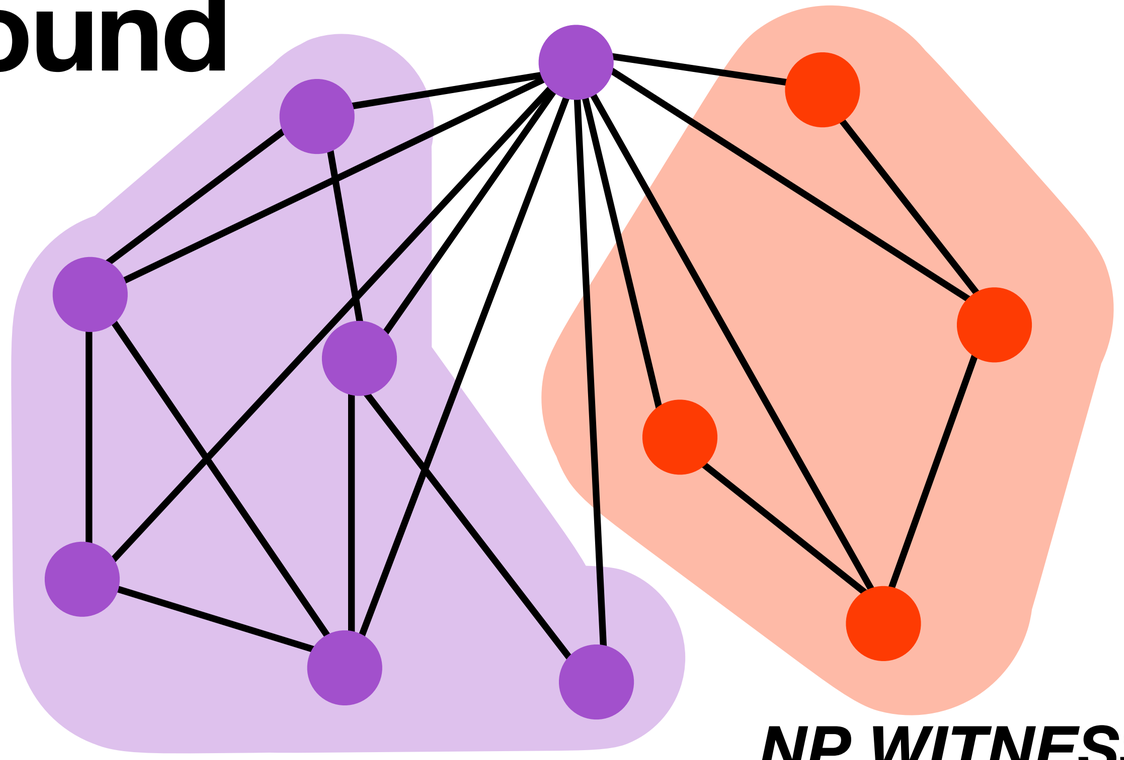
succinctness lower bound

CERTIFICATE



if this is asymptotically smaller than the witness, it is a SNARG for NP!

t corrupt and $n - t$ honest nodes



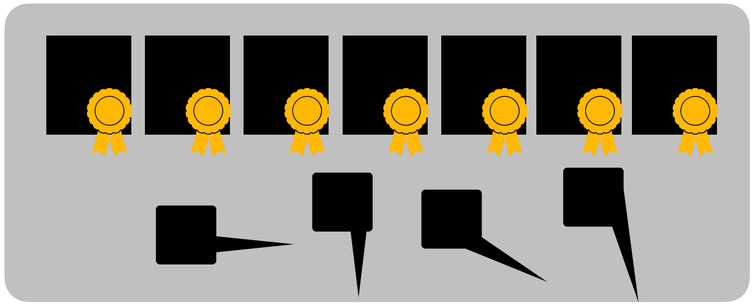
NP STATEMENT

NP WITNESS

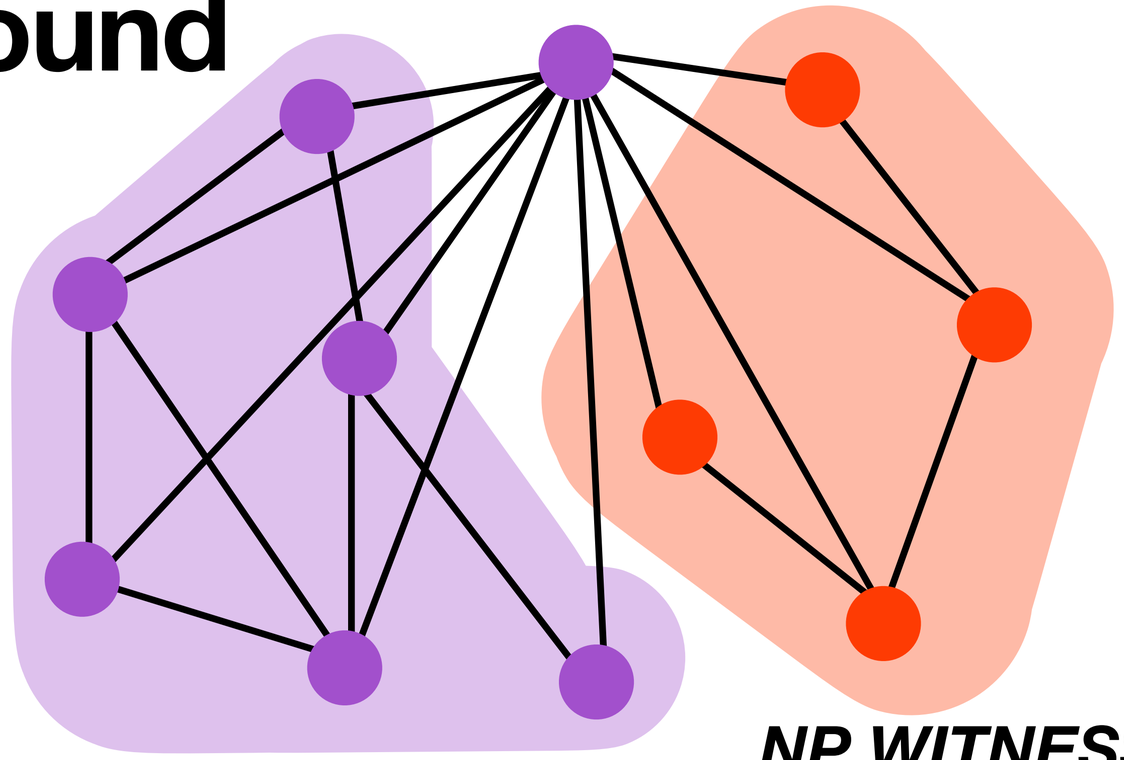
size $\binom{t}{2}$

succinctness lower bound

CERTIFICATE



if this is asymptotically smaller than the witness, it is a SNARG for NP!



NP STATEMENT

NP WITNESS

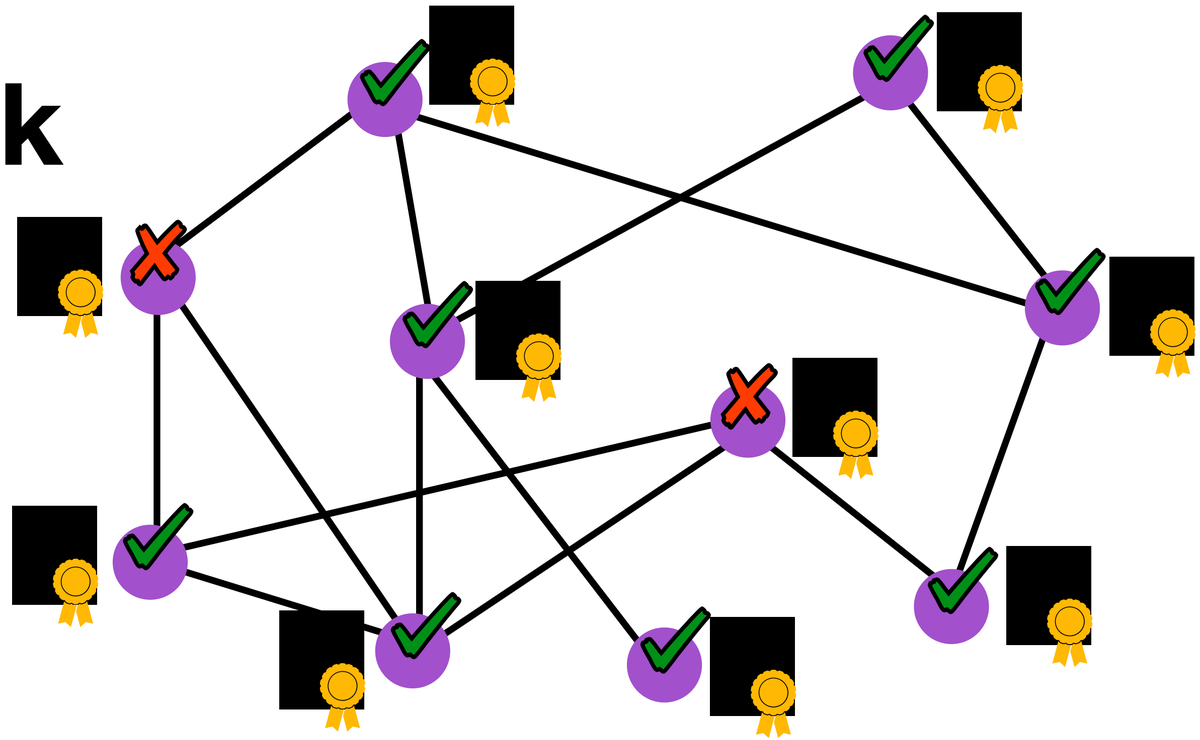
size $\binom{t}{2}$

t corrupt and $n - t$ honest nodes

we get a SNARG for NP if $\left|\ \right| \cdot (n-t) = o\left(\binom{t}{2}\right)$

distributed SNARGs on corrupted network

PROVER



COMPLETENESS all verifiers accept honestly generated certificates

SOUNDNESS if there is no satisfying network ***consistent with the honest view***, no efficient cheating prover can trick all honest verifiers into accepting

summary

1. we define a model for distributed SNARGs which are resilient to a partially corrupted network

n = network size, t = # of corrupt nodes

2. we construct a SNARG in this model for any property in P

with certificates of size $\tilde{O}\left(\binom{t}{2}/(n-t)\right)$ under LWE

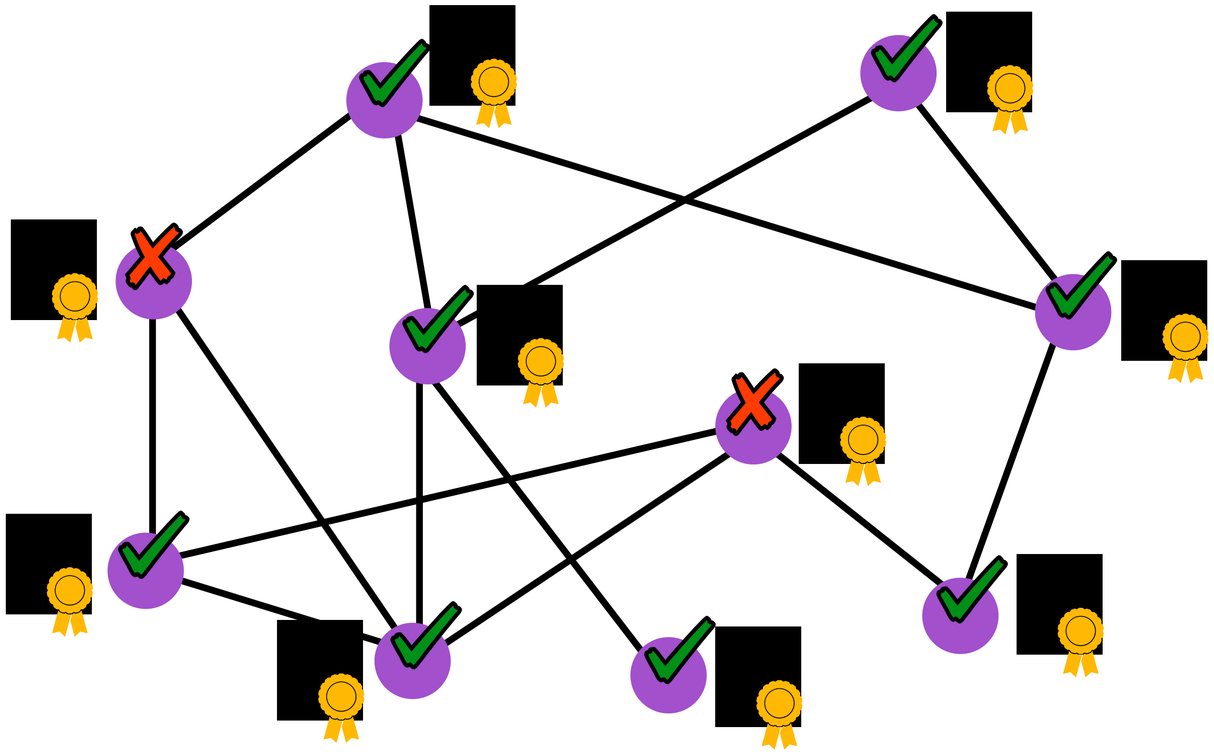
by using fully local SEH families and RAM SNARGs

3. we show that one would likely need heavier tools to get shorter certificates

open questions

- defining the model and building a construction for a *distributed* prover
- removing the assumptions about the network which are necessary for our current construction
- building a more succinct version of this object from a SNARG for NP (not a SNARK)

thank you!



full version: